

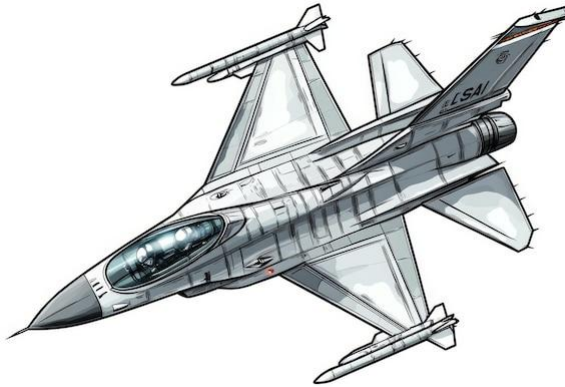
Control Barrier Functions: Theory Overview and Design Using Dynamic Safety Margins

Victor Freire

EuroGNC 2026, Madrid

May 4, 2026

Introduction



Introduction



Stability

Essentially solved

Introduction



Stability

Essentially solved

Safety

Under study
(Workshop)

Introduction

Constrained Control

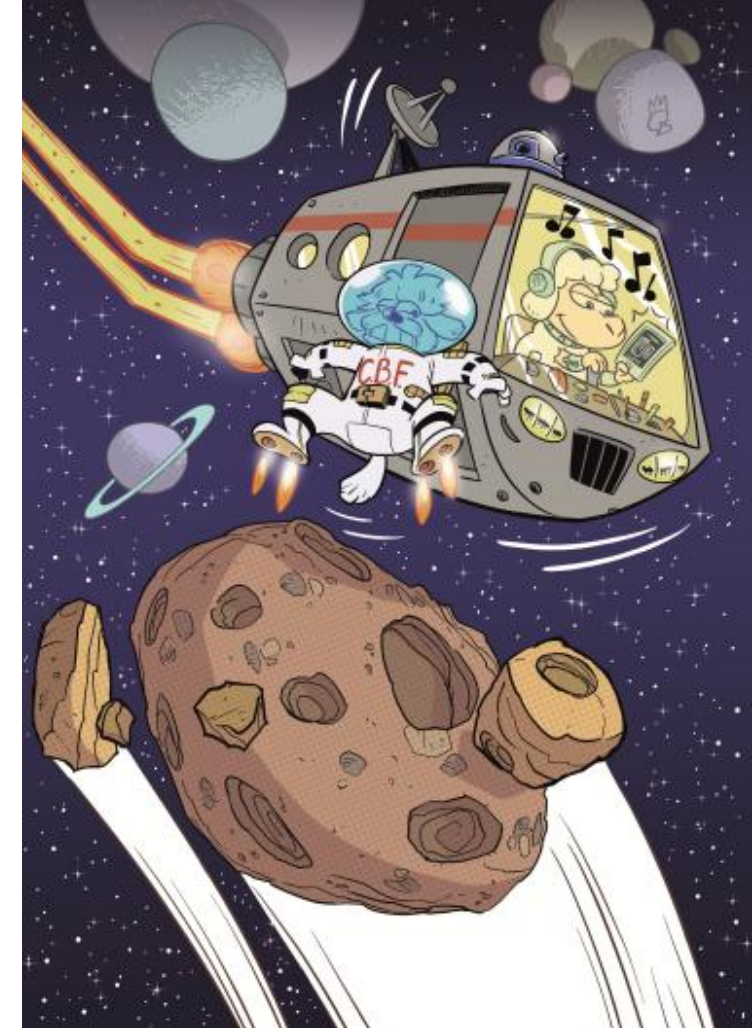
- **Model Predictive Control**
Full control law redesign
High computational cost



Introduction

Constrained Control

- **Model Predictive Control**
Full control law redesign
High computational cost
- **Safety Filters**
Keep legacy controller
Lower computational cost



Safety Filters

Family of constrained controllers that use a two-step approach:

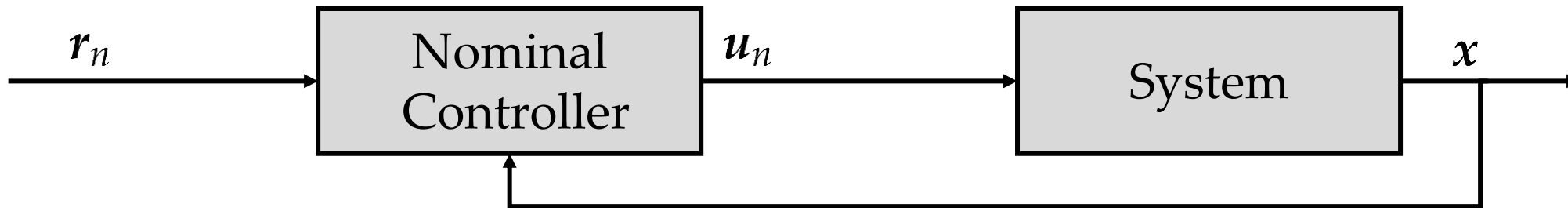


Introduction

Safety Filters

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1. Design a control law for the **Unconstrained** system,

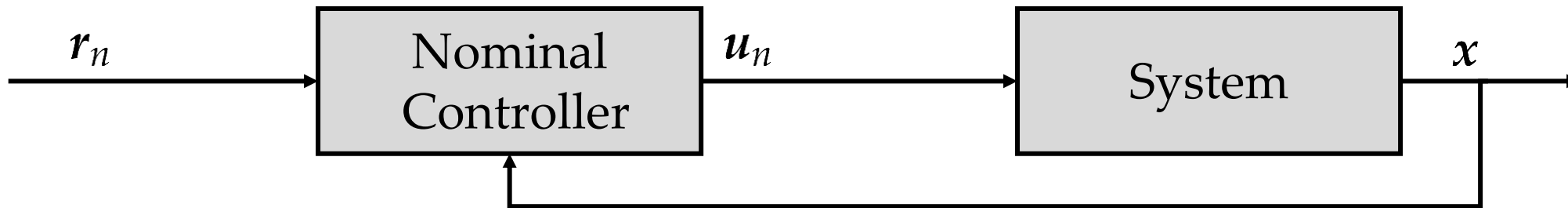


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Family of constrained controllers that use a two-step approach:

1. Design a control law for the **Unconstrained** system,
2. Introduce an **Add-on** unit for constraint enforcement.

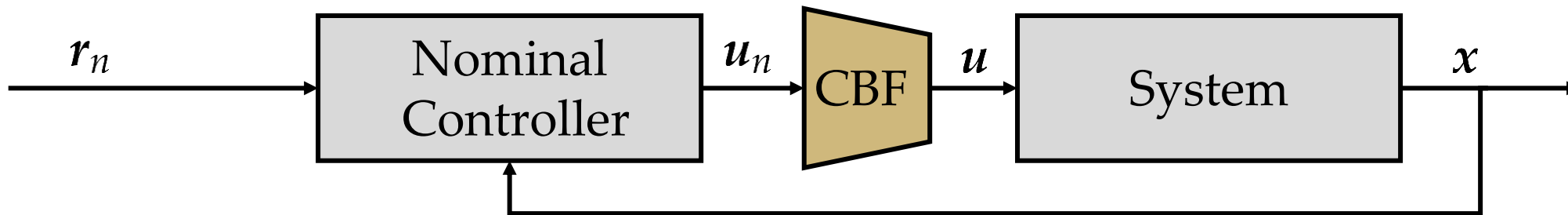


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Family of constrained controllers that use a two-step approach:

1. Design a control law for the **Unconstrained** system,
2. Introduce an **Add-on** unit for constraint enforcement.
 - **Control Barrier Function:** Filter nominal controller



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- ❖ Workshop Timeline
- ❖ Safety and Invariance
- ❖ Control Barrier Functions
- ❖ CBF-based Safety Filter
- ❖ Example: Safety Filter for F-16
- ❖ My Work: DSMs are CBFs



Safety and Invariance

Problem: Constrained Control



Safety and Invariance

Problem: Constrained Control

Given a control-affine system

$$\dot{x}(t) = f(x(t)) + g(x(t))u(t)$$



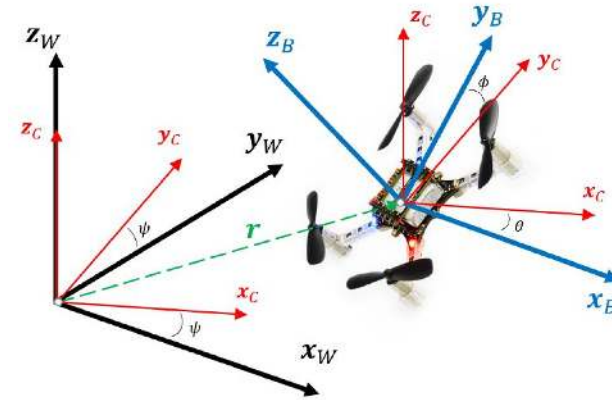
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Quadcopter Dynamics



$$x = \begin{bmatrix} r \\ \xi \\ \dot{r} \end{bmatrix} \quad \xi = \begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix}$$

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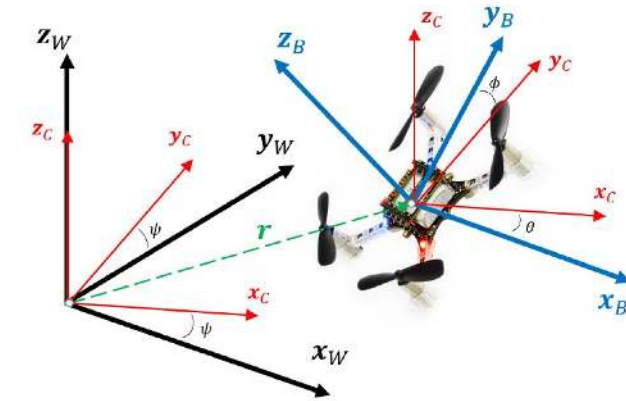
Quadcopter Dynamics

$$G_{\xi}(\xi) = \begin{bmatrix} 1 & s\phi t\theta & c\phi t\phi \\ 0 & c\phi & -s\phi \\ 0 & \frac{s\phi}{c\theta} & \frac{c\phi}{c\theta} \end{bmatrix}$$

$$G_T(\xi) = \begin{bmatrix} s\phi s\psi + c\phi s\theta c\psi \\ c\phi s\theta s\psi - s\phi c\psi \\ c\phi c\theta \end{bmatrix}$$

$$f(x) = \begin{bmatrix} \dot{r} \\ 0 \\ -g_a z_W \end{bmatrix}$$

$$g(x) = \begin{bmatrix} 0 & 0 \\ 0 & G_{\xi}(\xi) \\ G_T(\xi) & 0 \end{bmatrix}$$



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F-16 Pitch Dynamics

$$x = \begin{bmatrix} \theta & \dot{\theta} & \alpha_w \end{bmatrix}^\top$$

$$u = \begin{bmatrix} \delta_e & \delta_f \end{bmatrix}^\top$$



θ pitch α_w angle of attack δ_e, δ_f control surfaces

Safety and Invariance

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F-16 Pitch Dynamics

$$x = [\theta \quad \dot{\theta} \quad \alpha_w]^\top$$

$$u = [\delta_e \quad \delta_f]^\top$$

$$f(x) = Ax \quad g(x) = B$$



$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -0.87 & 43.22 \\ 0 & 0.99 & -1.34 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ -17.25 & -1.58 \\ -0.17 & -0.25 \end{bmatrix}$$

Short period approximation

Safety and Invariance

Problem: Constrained Control

Given a control-affine system

$$\dot{x}(t) = f(x(t)) + g(x(t))u(t)$$

and constraints

$$\mathcal{X} \subset \mathbb{R}^n, \mathcal{U} \subset \mathbb{R}^m,$$

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Pitch pointing constraint

$$\mathcal{X} = \{x \in \mathbb{R}^3 \mid |\alpha_w| \leq 4 \text{ deg}\}$$

Control surface deflection limits

$$\mathcal{U} = \{u \in \mathbb{R}^2 \mid |\delta_e| \leq 25 \text{ deg}, |\delta_f| \leq 20 \text{ deg}\}$$

Safety and Invariance

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Bonus: Trajectory $x(t)$ achieves goal.

(Performance)



Safety and Invariance

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Is there a solution?



Safety and Invariance

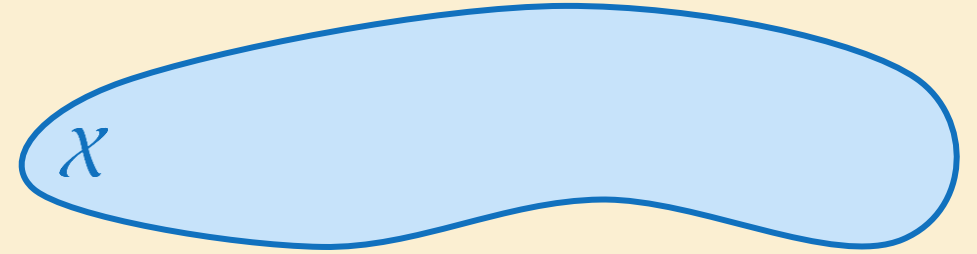
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$\mathcal{X}$ is **control invariant** if for any initial condition $x_0 \in \mathcal{X}$, there exists a signal $u : [0, \infty) \rightarrow \mathcal{U}$ such that $x(t) \in \mathcal{X}$.

Solution exists if $\mathcal{X}$ is **control invariant**.

Safety and Invariance

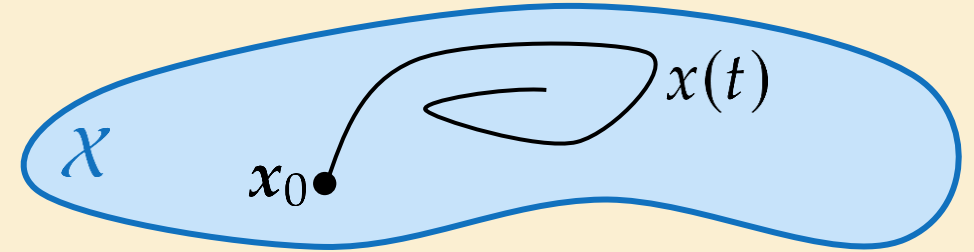
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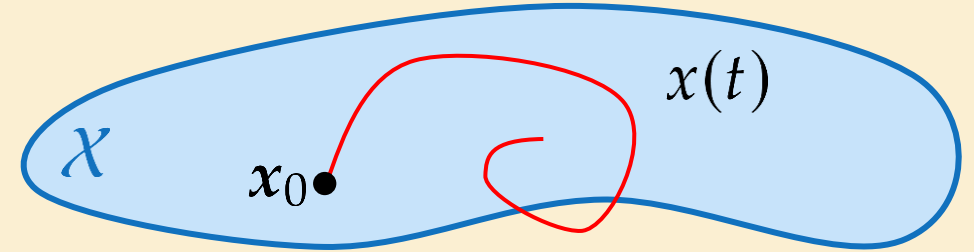
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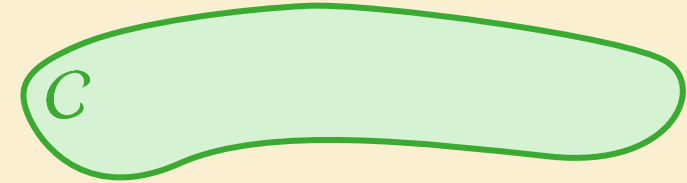
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$\mathcal{C}$ is **control invariant** if for any initial condition $x_0 \in \mathcal{C}$, there exists a signal $u : [0, \infty) \rightarrow \mathcal{U}$ such that $x(t) \in \mathcal{C}$.

Safety and Invariance

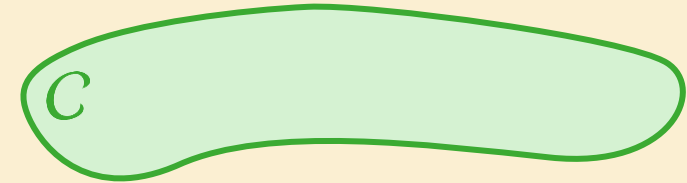
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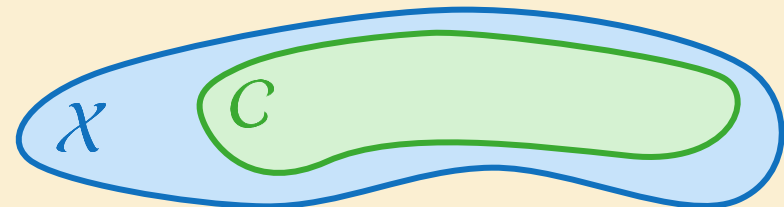
Solution exists if there exists $\mathcal{C}$ **control invariant** and **admissible** (i.e., safe).

Definition: Control Invariance



$\mathcal{C}$ is **control invariant** if for any initial condition $x_0 \in \mathcal{C}$, there exists a signal $u : [0, \infty) \rightarrow \mathcal{U}$ such that $x(t) \in \mathcal{C}$.

A set $\mathcal{C}$ is **admissible** if $\mathcal{C} \subset \mathcal{X}$.



Safety and Invariance

Theorem: Nagumo (simplified)

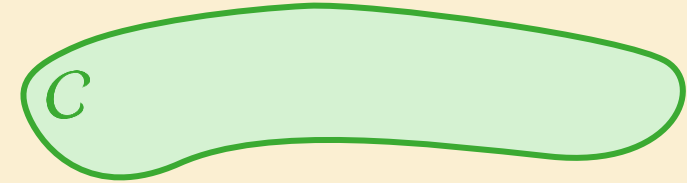
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$$\mathcal{C} = \{x \in \mathbb{R}^n \mid h(x) \geq 0\},$$

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$$\forall x \in \partial \mathcal{C}, \exists u \in \mathcal{U}, L_f h(x) + L_g h(x)u \geq 0.$$

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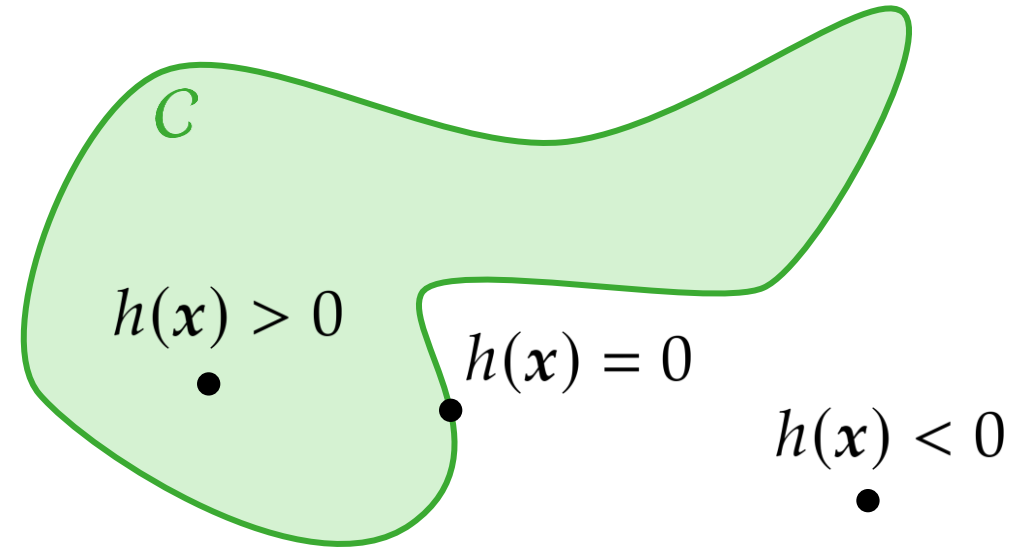
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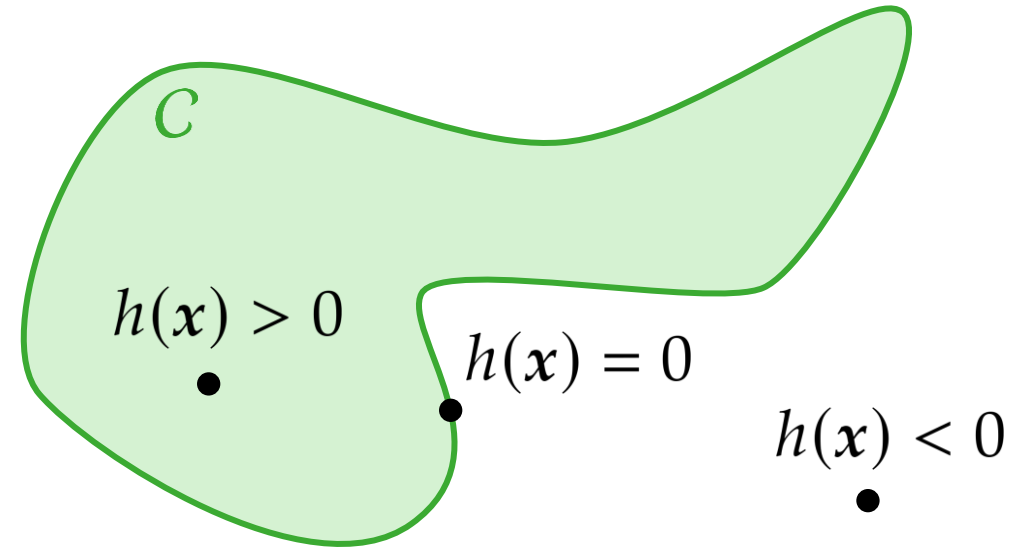
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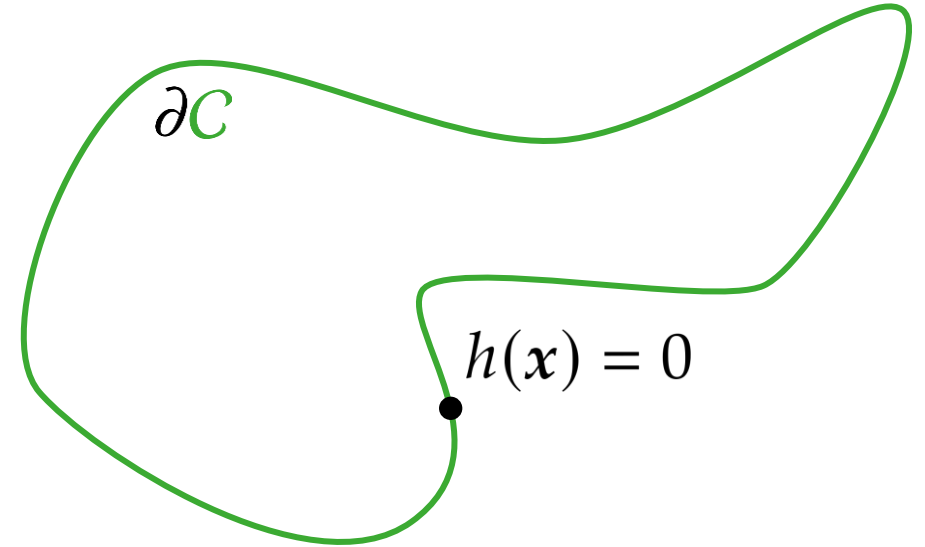
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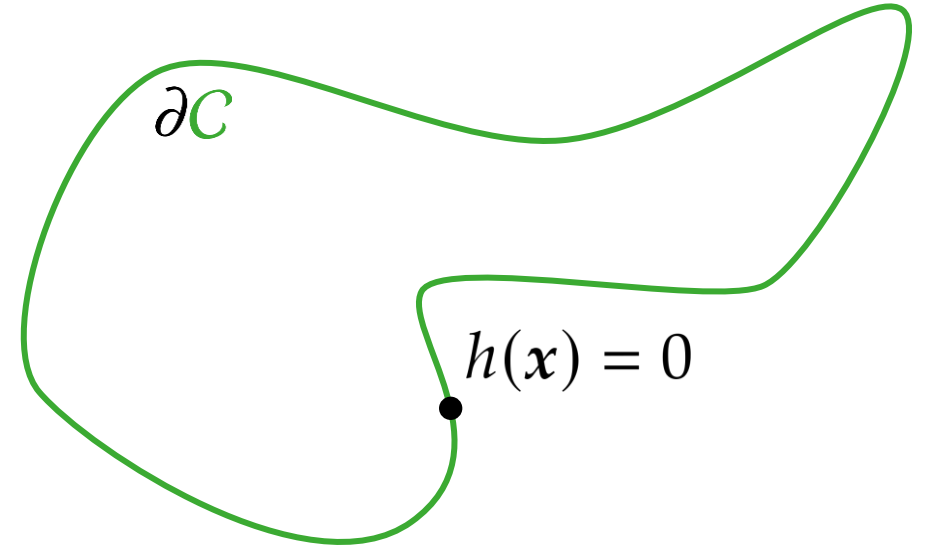
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$$L_f h(x) + L_g h(x)u$$



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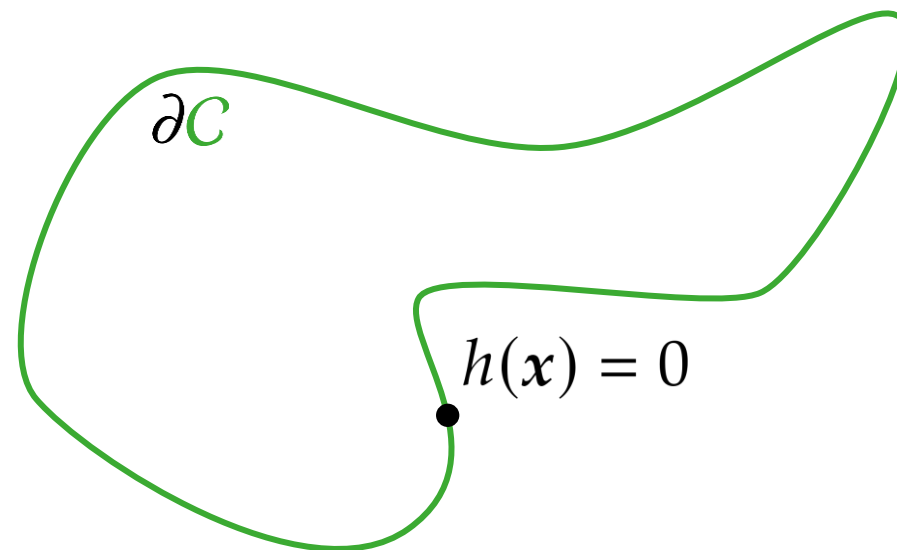
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“Lie derivatives” (Notation)

$$L_f h(x) \triangleq \frac{\partial h}{\partial x} f(x) \quad L_g h(x) \triangleq \frac{\partial h}{\partial x} g(x)$$



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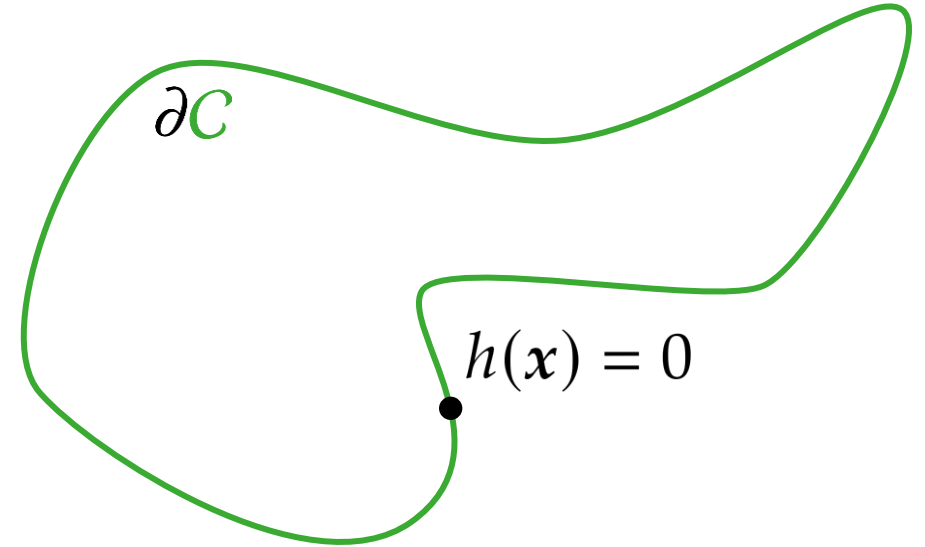
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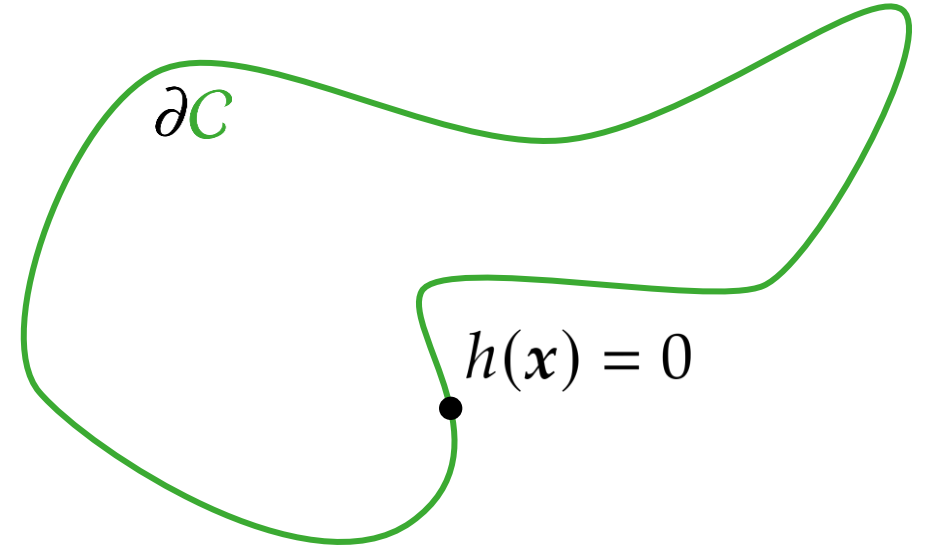
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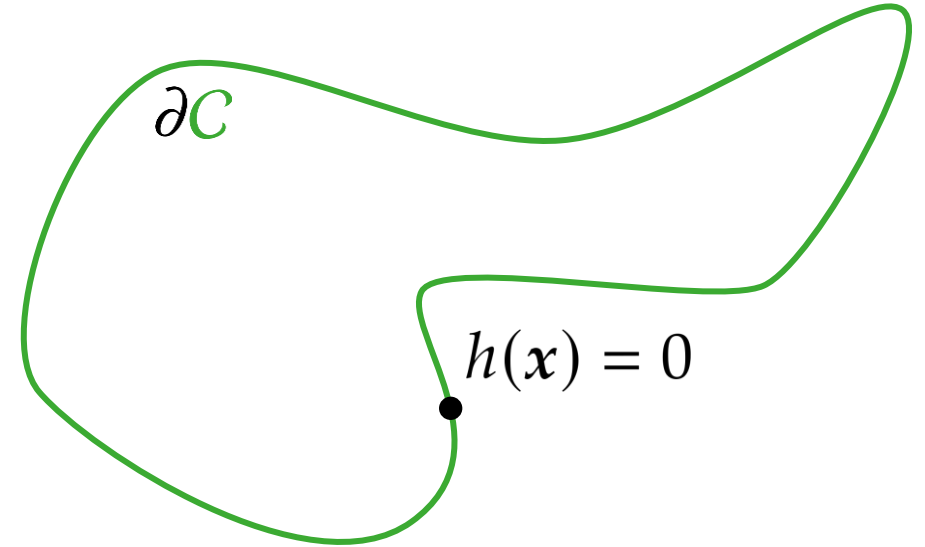
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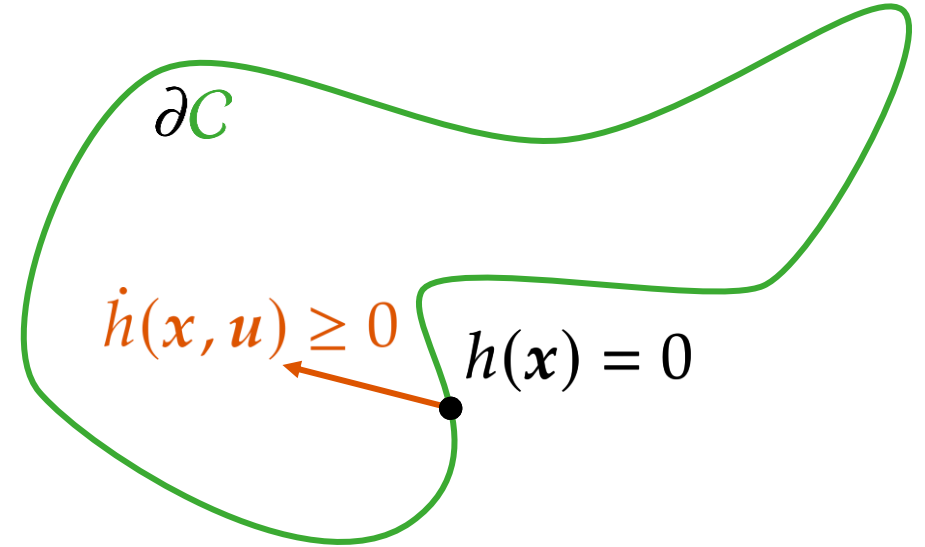
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Safety and Invariance

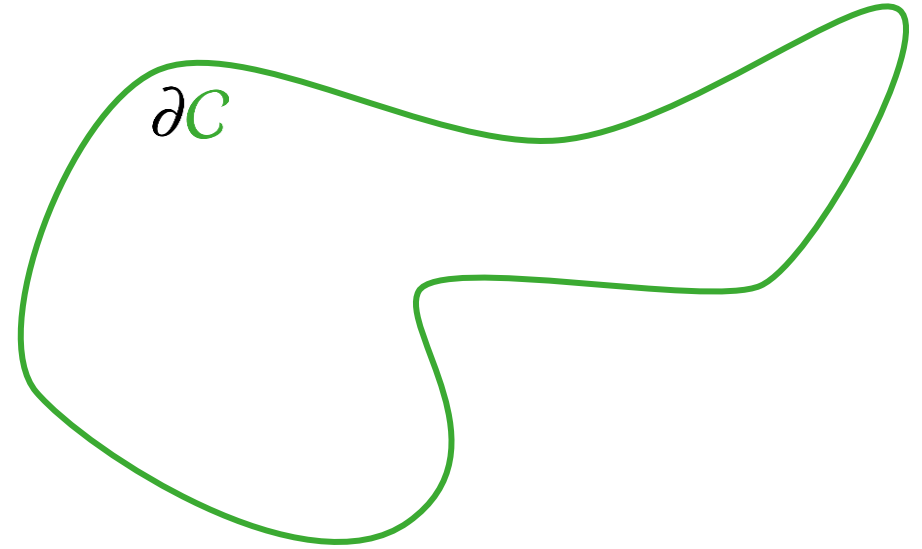
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Measure-zero.

Difficult to check numerically.

Safety and Invariance

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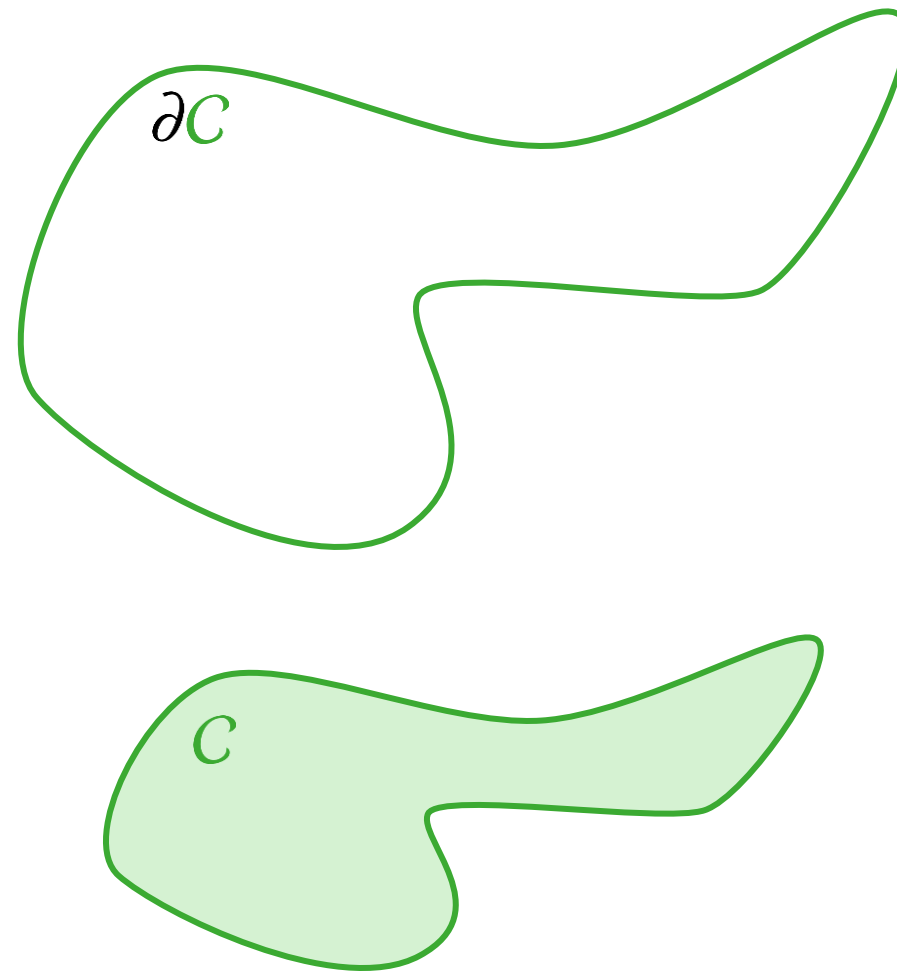
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Theorem: CBF Condition (2017)

$$\forall x \in \mathcal{C}, \exists u \in \mathcal{U}, L_f h(x) + L_g h(x)u + \alpha(h(x)) \geq 0.$$



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Control Barrier Functions (CBFs)

Definition: CBF



Ames, Xu, Grizzle, Tabuada, *TAC* 2017

Ames, Coogan, Egerstedt, Notomista, Sreenath, Tabuada, *ECC* 2019 (survey)

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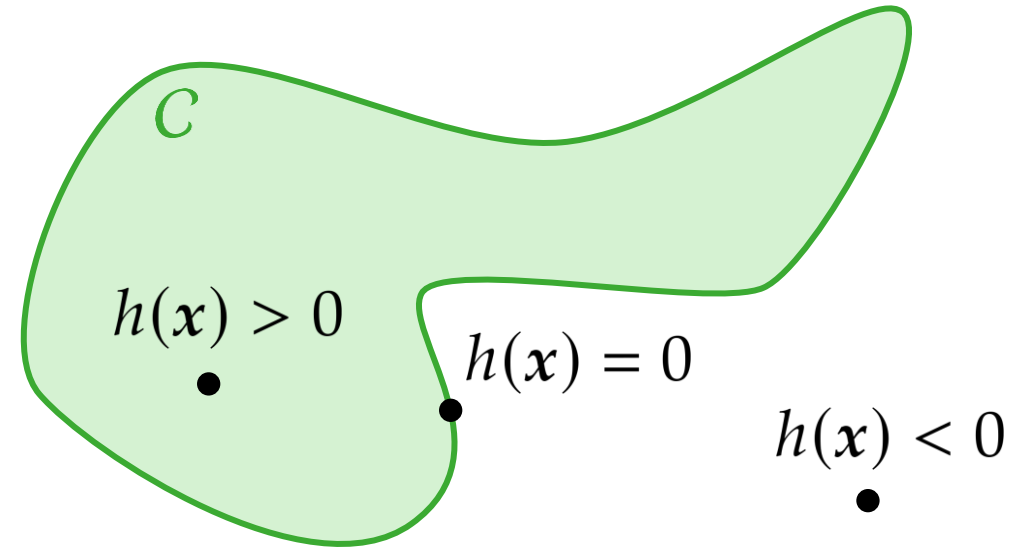
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Control Barrier Functions (CBFs)

Definition: CBF

Candidate CBF

A smooth function $h : \mathbb{R}^n \rightarrow \mathbb{R}$
such that $\mathcal{C} = \{x \in \mathbb{R}^n \mid h(x) \geq 0\}$



Control Barrier Functions (CBFs)

Definition: CBF

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CBF Condition

There exists $\alpha \in \mathcal{K}$ such that

$$\forall x \in \mathcal{C}, \exists u \in \mathcal{U}, \dot{h}(x, u) + \alpha(h(x)) \geq 0,$$

where $\dot{h}(x, u) = L_f h(x) + L_g h(x)u$.



Control Barrier Functions (CBFs)

Definition: CBF

Candidate CBF

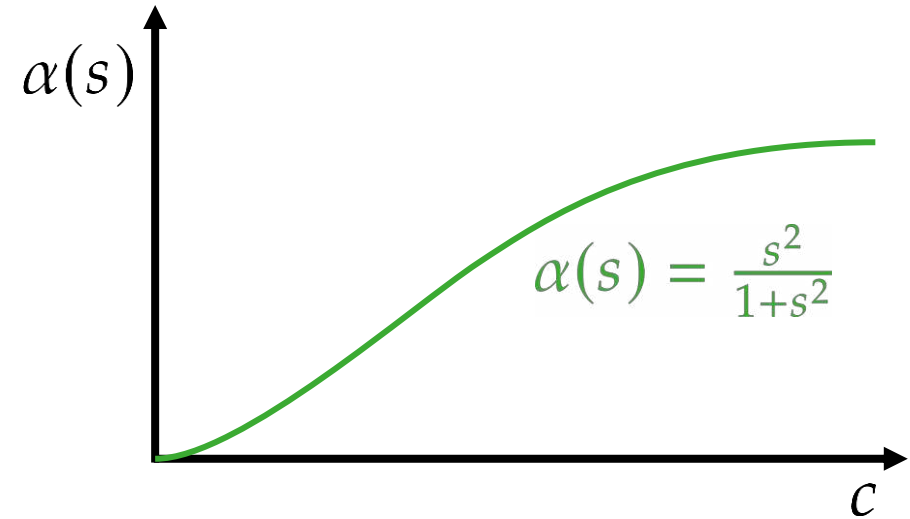
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CBF Condition

There exists $\alpha \in \mathcal{K}$ such that

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where $\dot{h}(x, u) = L_f h(x) + L_g h(x)u$.



$\alpha \in \mathcal{K}$ if

- continuous
- strictly increasing
- $\alpha(0) = 0$



Control Barrier Functions (CBFs)

Definition: CBF

Candidate CBF

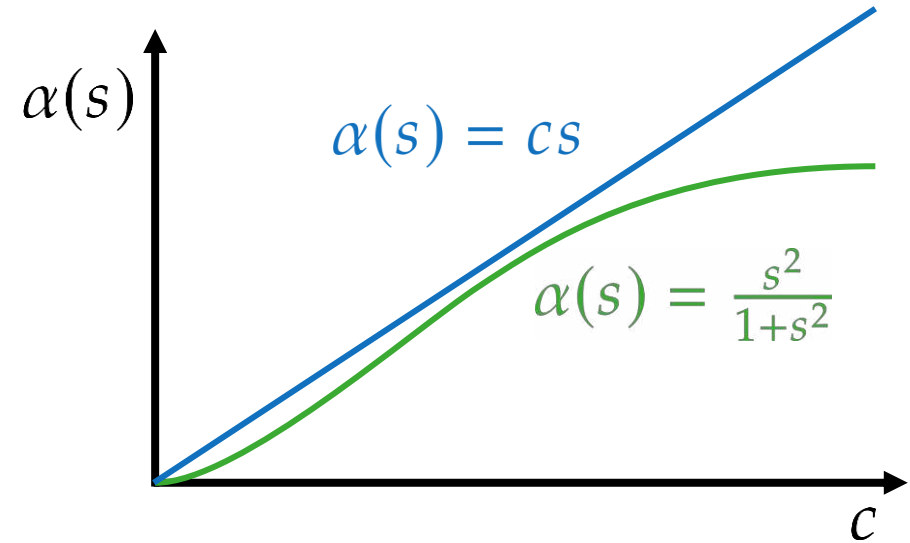
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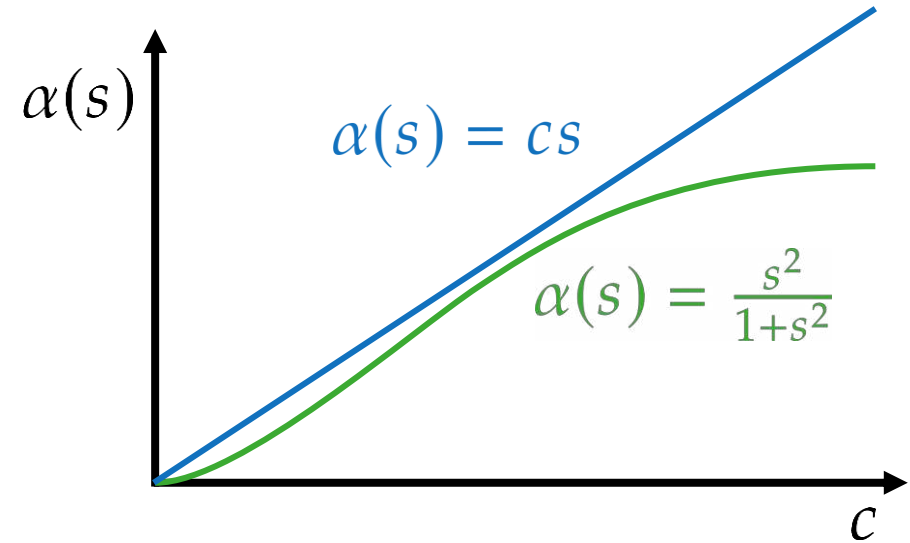
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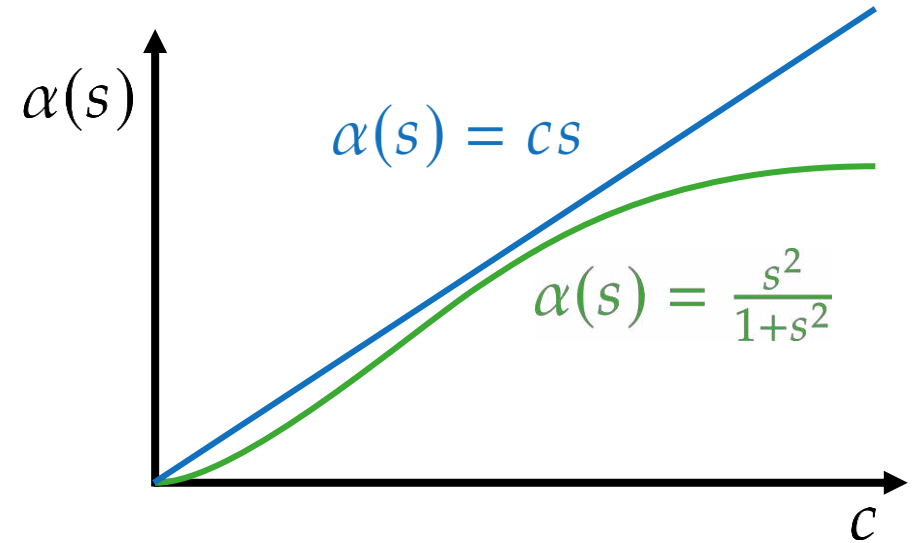
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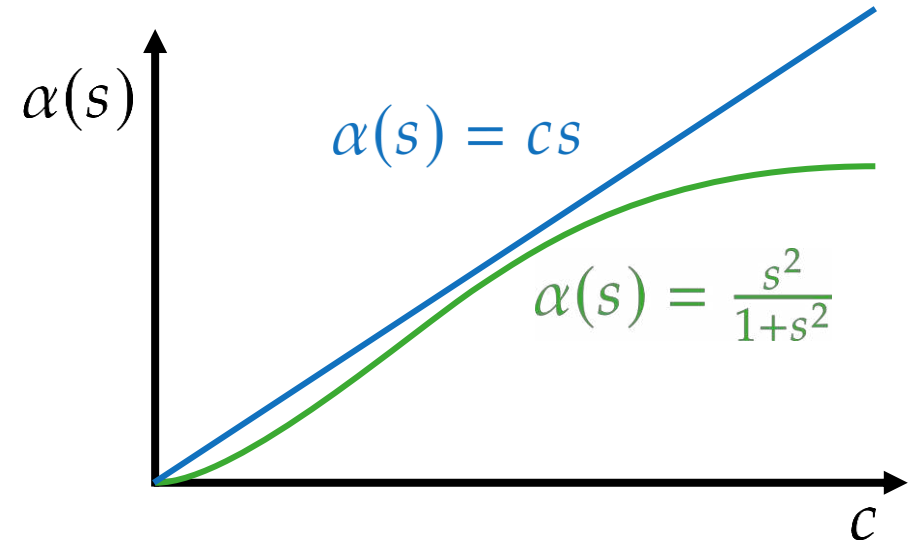
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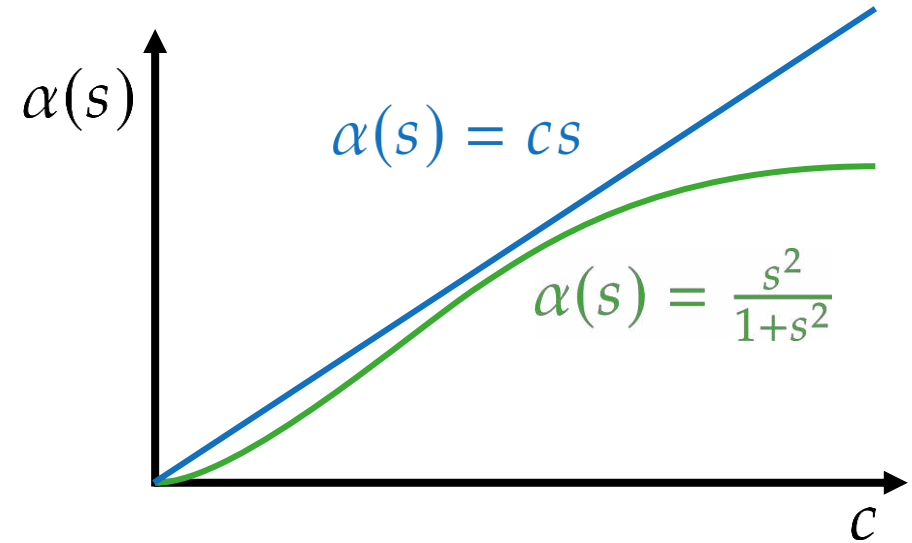
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Theorem: Nagumo (simplified)

Let $h : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuously differentiable with regular value 0. The set

$$\mathcal{C} = \{x \in \mathbb{R}^n \mid h(x) \geq 0\},$$

is **control invariant** if and only if

$$\forall x \in \partial \mathcal{C}, \exists u \in \mathcal{U}, L_f h(x) + L_g h(x)u \geq 0.$$

$$x \in \partial \mathcal{C} \implies h(x) = 0 \implies \alpha(h(x)) = 0$$



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Theorem: CBF Condition (2017)

If h is a CBF, then $\mathcal{C}$ is control invariant.



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Theorem: CBF Condition (2017)

If h is a CBF, then $\mathcal{C}$ is control invariant.

What about the other direction?

If $\mathcal{C}$ is control invariant, then h is a CBF.



Control Barrier Functions (CBFs)

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Theorem: CBF Condition (2017)

If h is a CBF, then $\mathcal{C}$ is control invariant.

Theorem: CBF Condition (2017)

Assume (a) $\mathcal{C}$ is compact and (b) there exists a continuous selection $\mu : \mathcal{C} \rightarrow \mathcal{U}$ such that $\mu(x) \in \mathcal{K}(x)$.

If $\mathcal{C}$ is control invariant, then h is a CBF.

$$\mathcal{K}(x) = \{u \in \mathcal{U} \mid L_f h(x) + L_g h(x)u \geq 0\}$$



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Problem: Constrained Control

Given a control-affine system
 $\dot{x}(t) = f(x(t)) + g(x(t))u(t)$
and constraints

$\mathcal{X} \subset \mathbb{R}^n, \mathcal{U} \subset \mathbb{R}^m,$
find a signal $u : [0, \infty) \rightarrow \mathcal{U}$
such that $x(t) \in \mathcal{X}$ for all $t \geq 0$.

Solution exists if there exists $\mathcal{C}$ **control invariant** and **admissible** (i.e., **safe**).



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Given a CBF h , how to find a solution $u(t)$?



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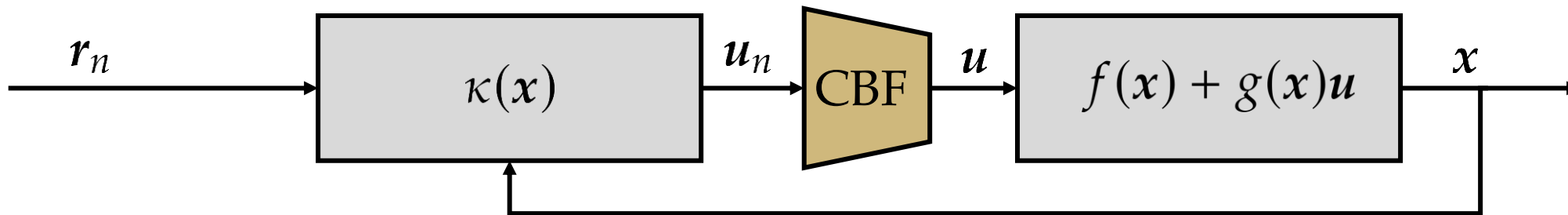


CBF-based Safety Filter

Safety Filters

Family of constrained controllers that use a two-step approach:

1. Design a control law for the **Unconstrained** system,
2. Introduce an **Add-on** unit for constraint enforcement.
 - **Control Barrier Function:** Filter nominal controller



CBF-based Safety Filter

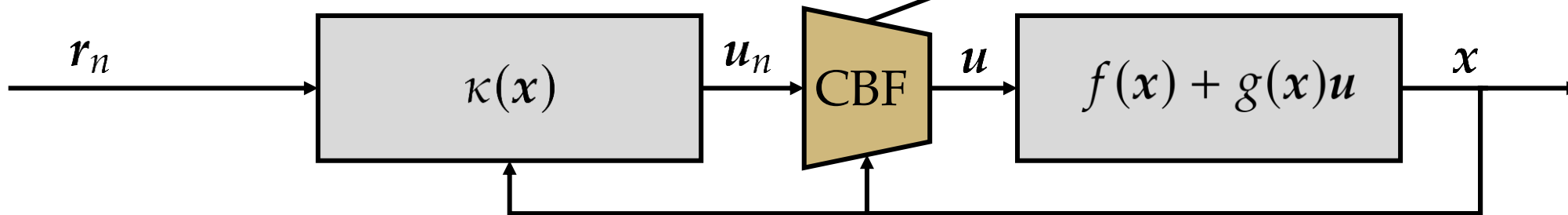
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$$\begin{aligned} \min_{u \in \mathcal{U}} \quad & \|u - \kappa(x)\|^2 \\ \text{s.t.} \quad & \dot{h}(x, u) + \alpha(h(x)) \geq 0 \end{aligned}$$



CBF-based Safety Filter

$$\begin{aligned} \min_{u \in \mathcal{U}} \quad & \|u - \kappa(x)\|^2 \\ \text{s.t.} \quad & L_f h(x) + L_g h(x)u \geq -\alpha(h(x)) \end{aligned}$$



CBF-based Safety Filter

Optimization Problem (Program)

$$\begin{aligned} \min_{\mathbf{u} \in \mathcal{U}} \quad & \|\mathbf{u} - \kappa(\mathbf{x})\|^2 \\ \text{s.t.} \quad & L_f h(\mathbf{x}) + L_g h(\mathbf{x})\mathbf{u} \geq -\alpha(h(\mathbf{x})) \end{aligned}$$



Find the vector $\mathbf{u} \in \mathcal{U}$ that minimizes $\|\mathbf{u} - \kappa(\mathbf{x})\|^2$ subject to $L_f h(\mathbf{x}) + L_g h(\mathbf{x})\mathbf{u} \geq -\alpha(h(\mathbf{x}))$.



CBF-based Safety Filter

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=

Find the vector $u \in \mathcal{U}$ that minimizes $\|u - \kappa(x)\|^2$ subject to $L_f h(x) + L_g h(x)u \geq -\alpha(h(x))$.

Given x , these **terms** are constant.



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If $\mathcal{U}$ is polyhedral $\mathcal{U} = \{u \mid Mu \leq b\}$,
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Commercial QP Solvers

mosek

GUROBI
OPTIMIZATION

COPT
Cardinal Optimizer

IBM
CPLEX



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Optimization Problem

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Is there a solution?

(feasibility)



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Theorem: CBF Safety Filter

If h is a CBF, then the CBF-based safety filter is feasible for all $x \in \mathcal{C}$.



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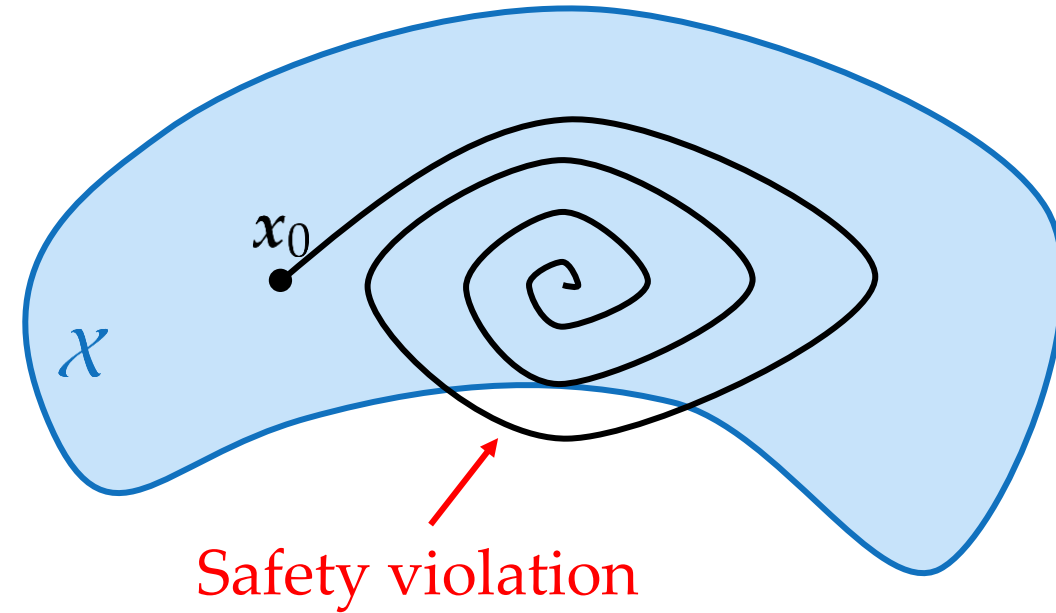
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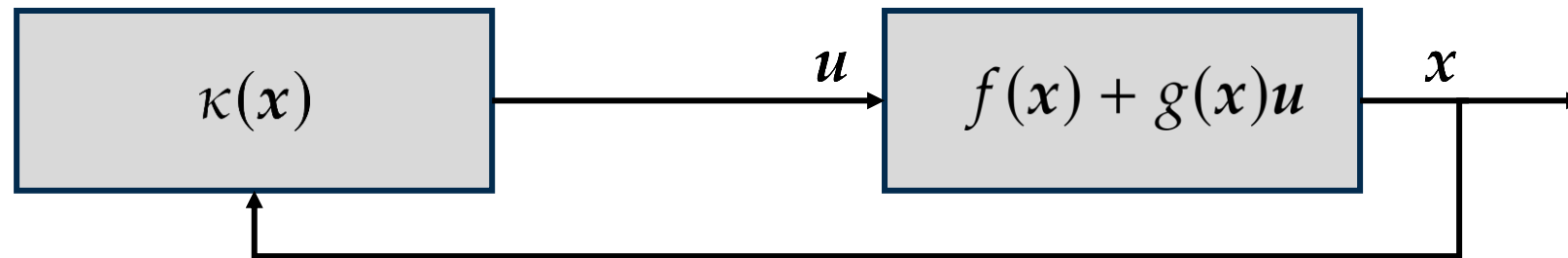
CBF-based Safety Filter

Safety Filters

1. Design a control law for the **Unconstrained** system,



Unsafe



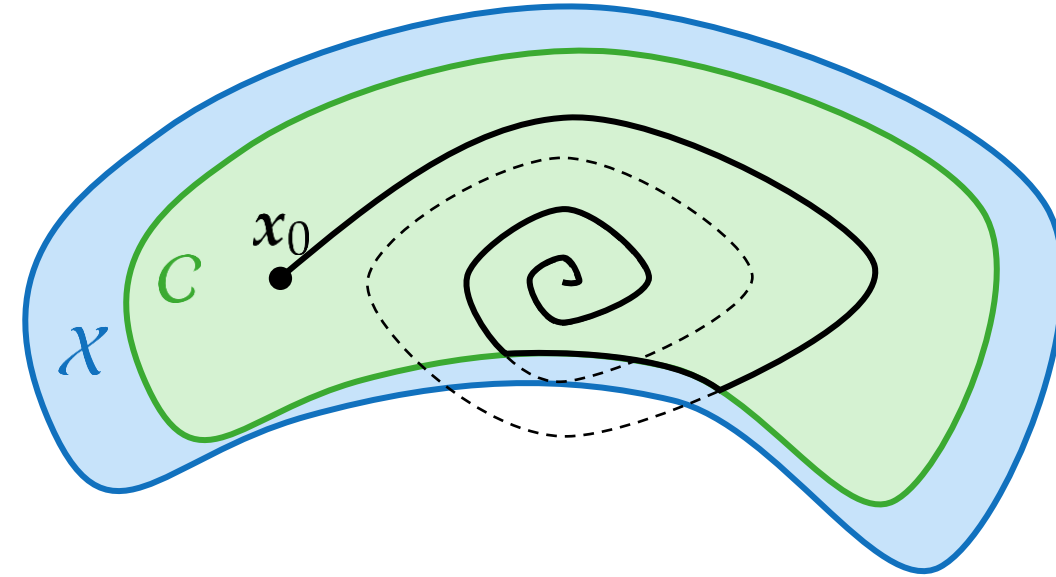
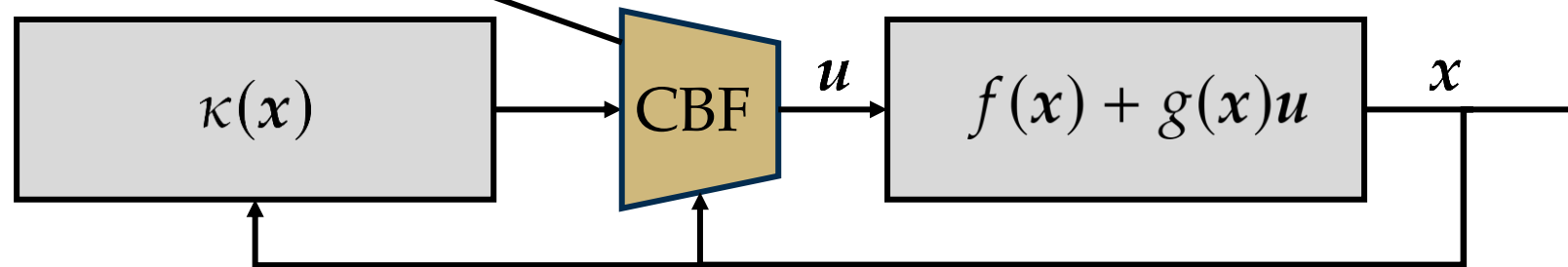
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Safe

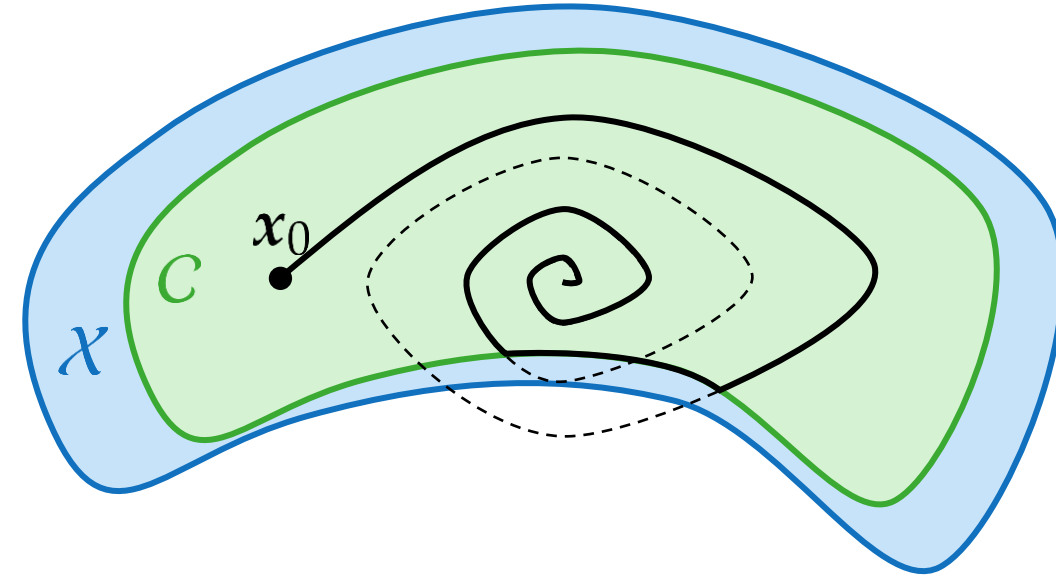


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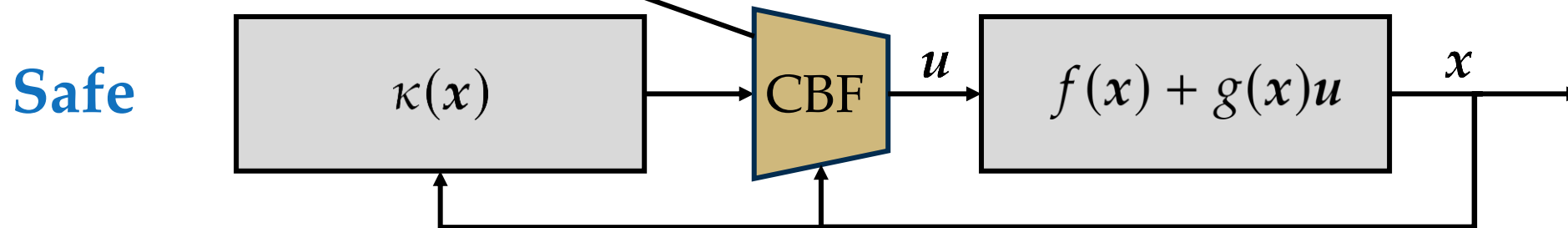
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How to find a CBF $h(x)$?



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Example: Safety Filter for F-16

F-16 Pitch Dynamics

$$\mathbf{x} = \begin{bmatrix} \theta & \dot{\theta} & \alpha_w \end{bmatrix}$$

$$\mathbf{u} = \begin{bmatrix} \delta_e & \delta_f \end{bmatrix}$$

$$\mathbf{f}(\mathbf{x}) = \mathbf{A}\mathbf{x} \quad \mathbf{g}(\mathbf{x}) = \mathbf{B}$$



$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -0.87 & 43.22 \\ 0 & 0.99 & -1.34 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ -17.25 & -1.58 \\ -0.17 & -0.25 \end{bmatrix}$$

θ pitch α_w angle of attack δ_e, δ_f control surfaces

Example: Safety Filter for F-16

F-16 Pitch Dynamics

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$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$

$$f(\mathbf{x}) = A\mathbf{x} \quad g(\mathbf{x}) = B$$



$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & -0.87 & 43.22 & -17.25 & -1.58 \\ 0 & 0.99 & -1.34 & -0.17 & -0.25 \\ 0 & 0 & 0 & -20 & 0 \\ 0 & 0 & 0 & 0 & -20 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 20 & 0 \\ 0 & 20 \end{bmatrix}$$

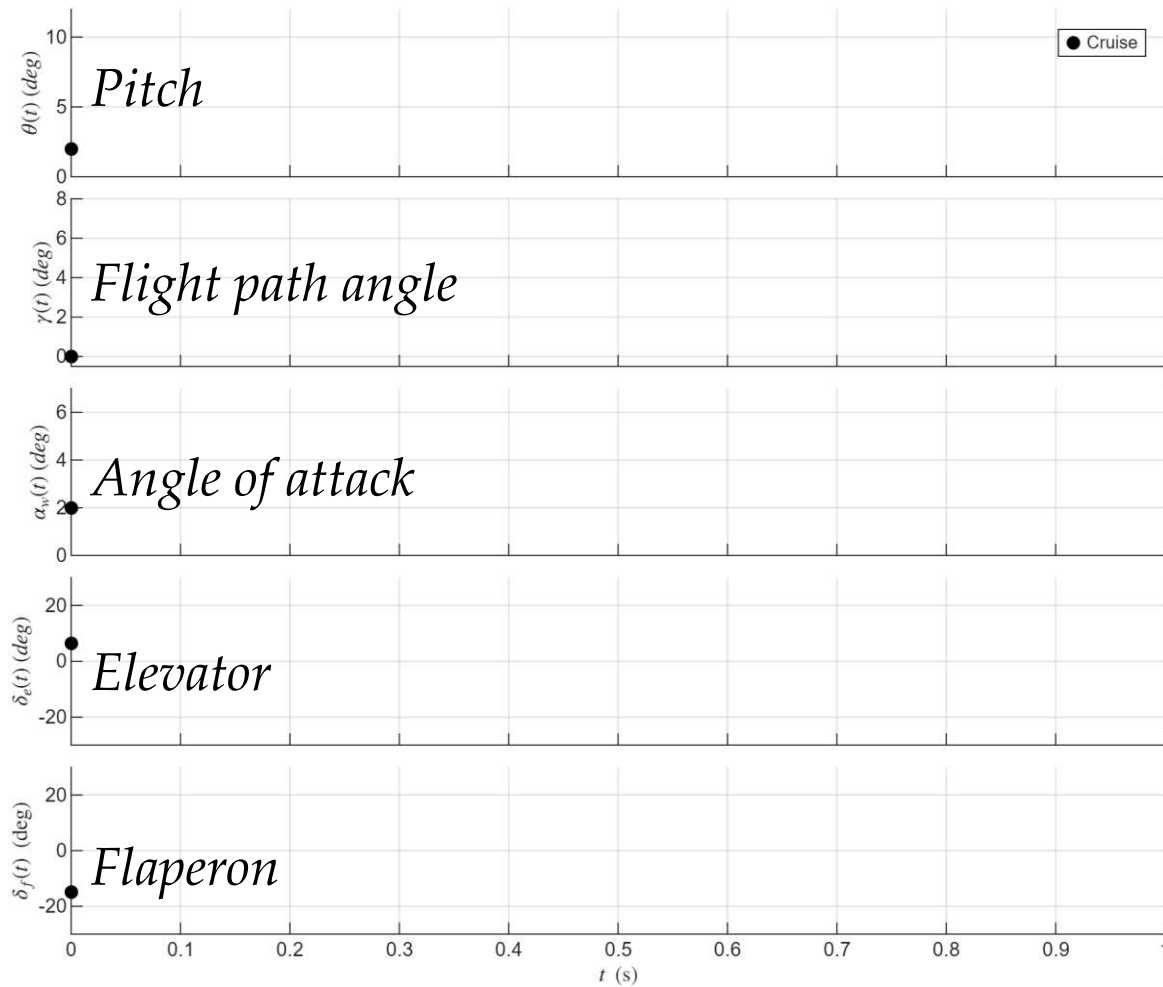
**Actuator Inner-loop
(first-order lag)**

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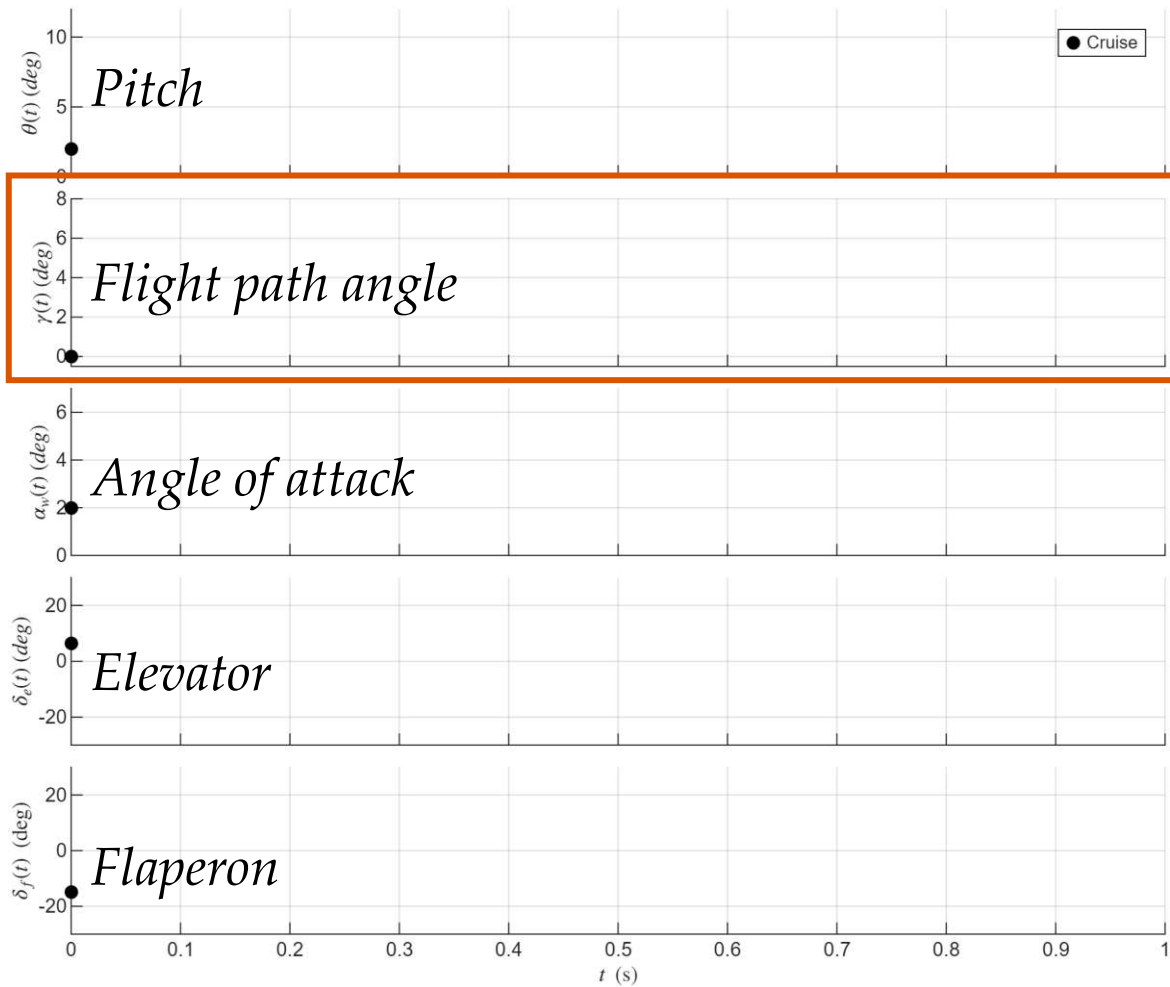
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Cruise: $\theta = 2, \gamma = 0$



Example: Safety Filter for F-16



F-16 Pitch Dynamics

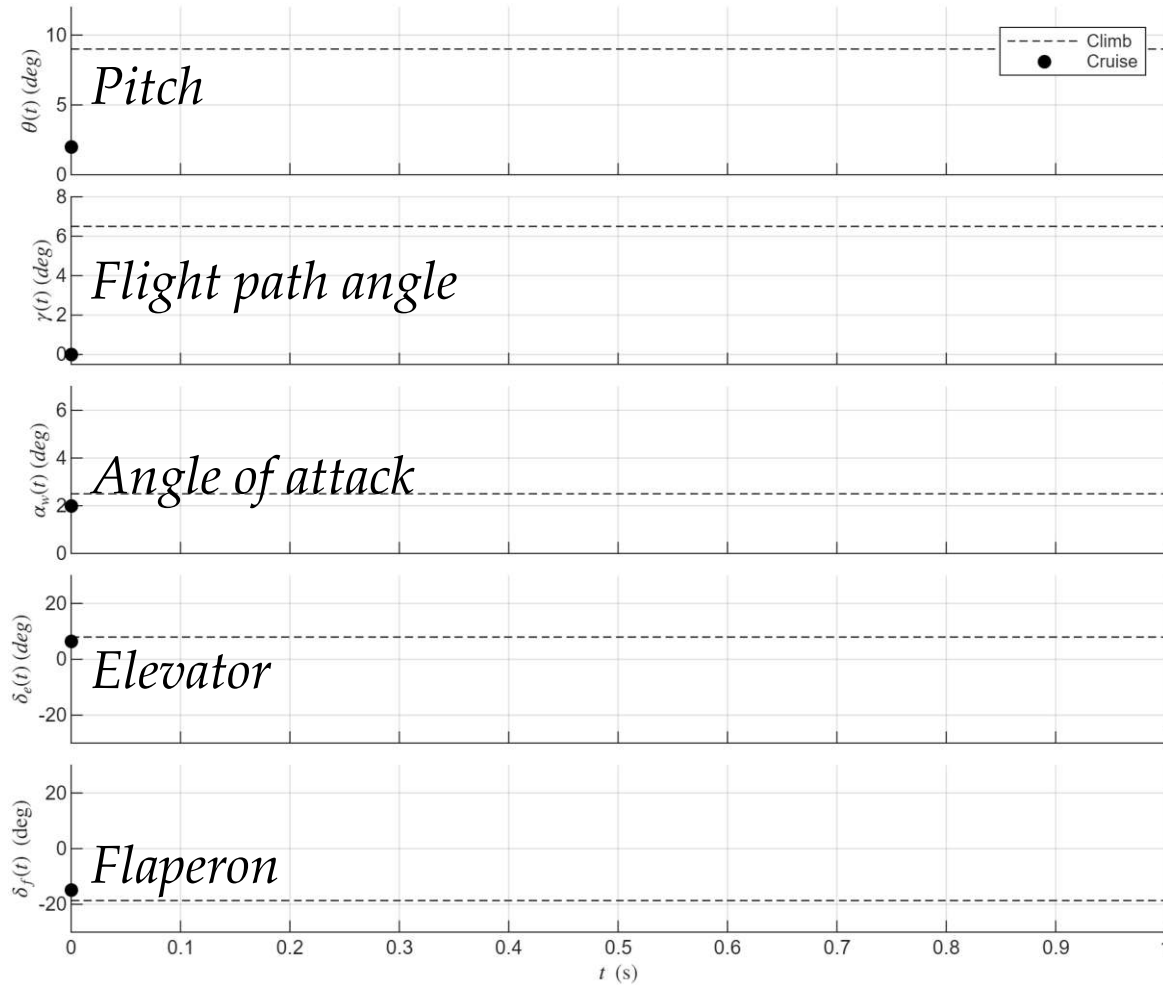
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Cruise: $\theta = 2$, $\gamma = 0$ $\gamma \approx \theta - \alpha_w$

Example: Safety Filter for F-16



F-16 Pitch Dynamics

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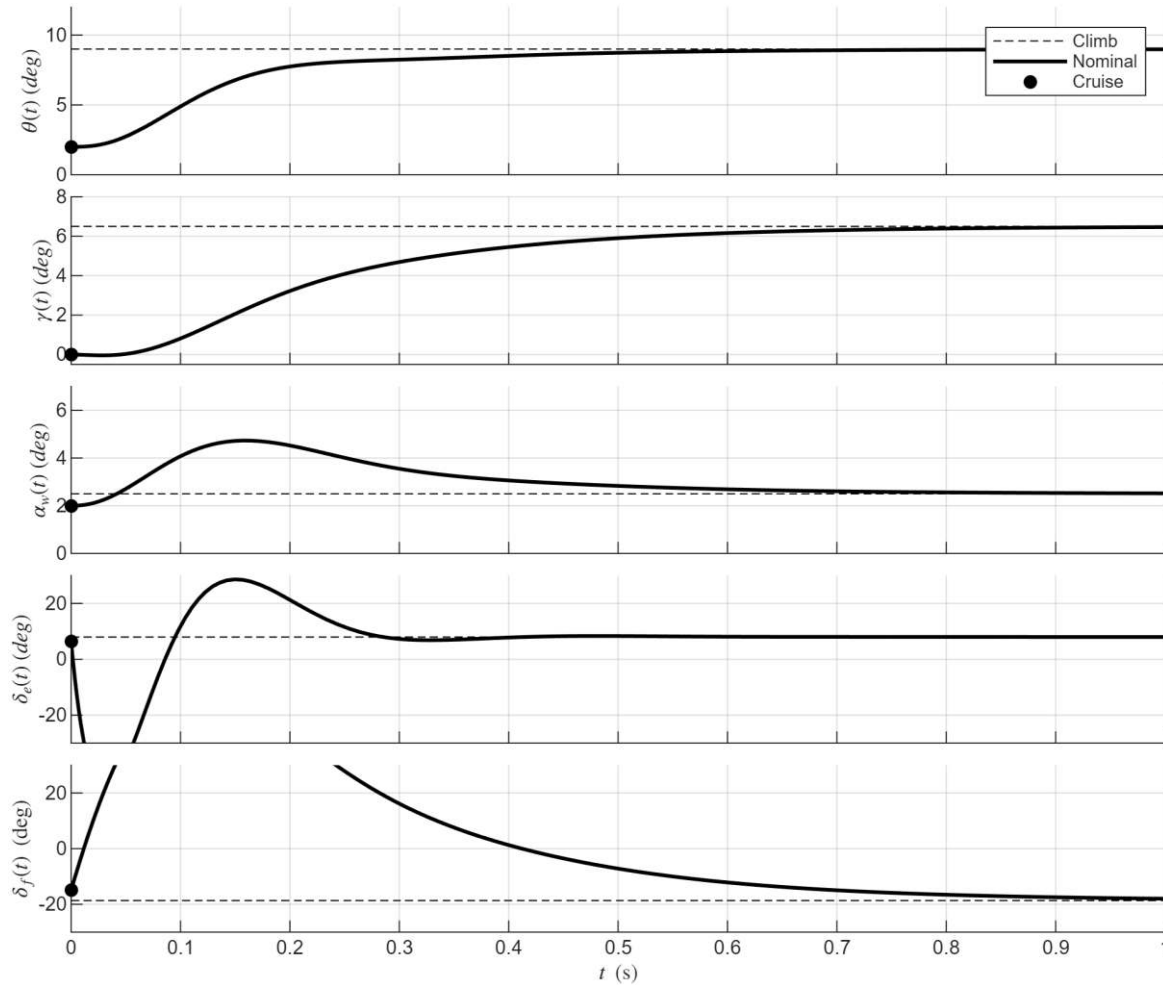
Cruise: $\theta = 2, \gamma = 0$



Goal

Climb: $\theta = 9, \gamma = 6.5$

Example: Safety Filter for F-16



F-16 Pitch Dynamics

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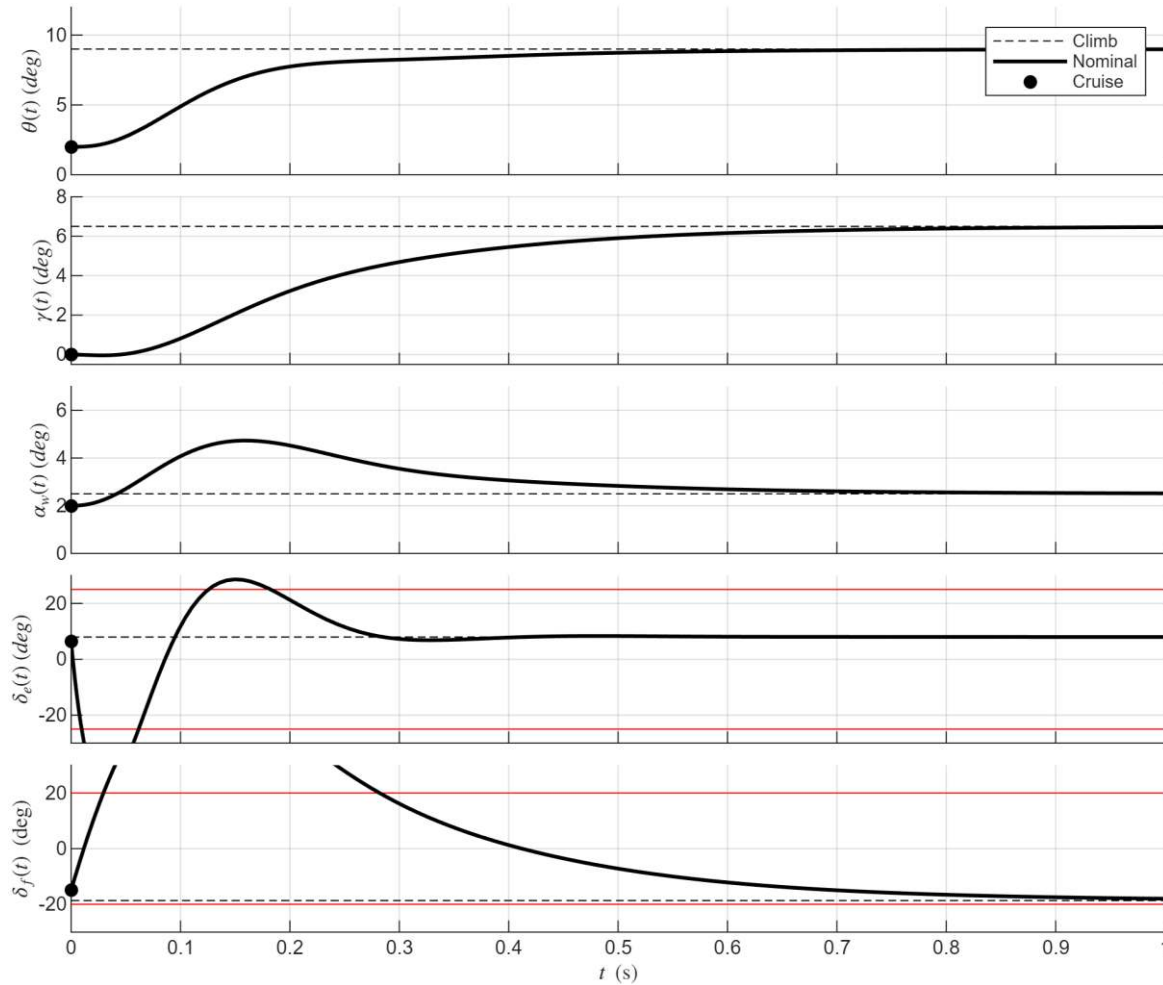
$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$



LQR Controller $\kappa(\mathbf{x}) = \bar{\mathbf{u}}(\mathbf{r}) - \mathbf{K}(\mathbf{x} - \bar{\mathbf{x}}(\mathbf{r}))$

Desired **Climb**
operating conditions

Example: Safety Filter for F-16



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Control surface deflection limits
 $|\delta_e| \leq 25$ deg, $|\delta_f| \leq 20$ deg

Example: Safety Filter for F-16

F-16 Pitch Dynamics

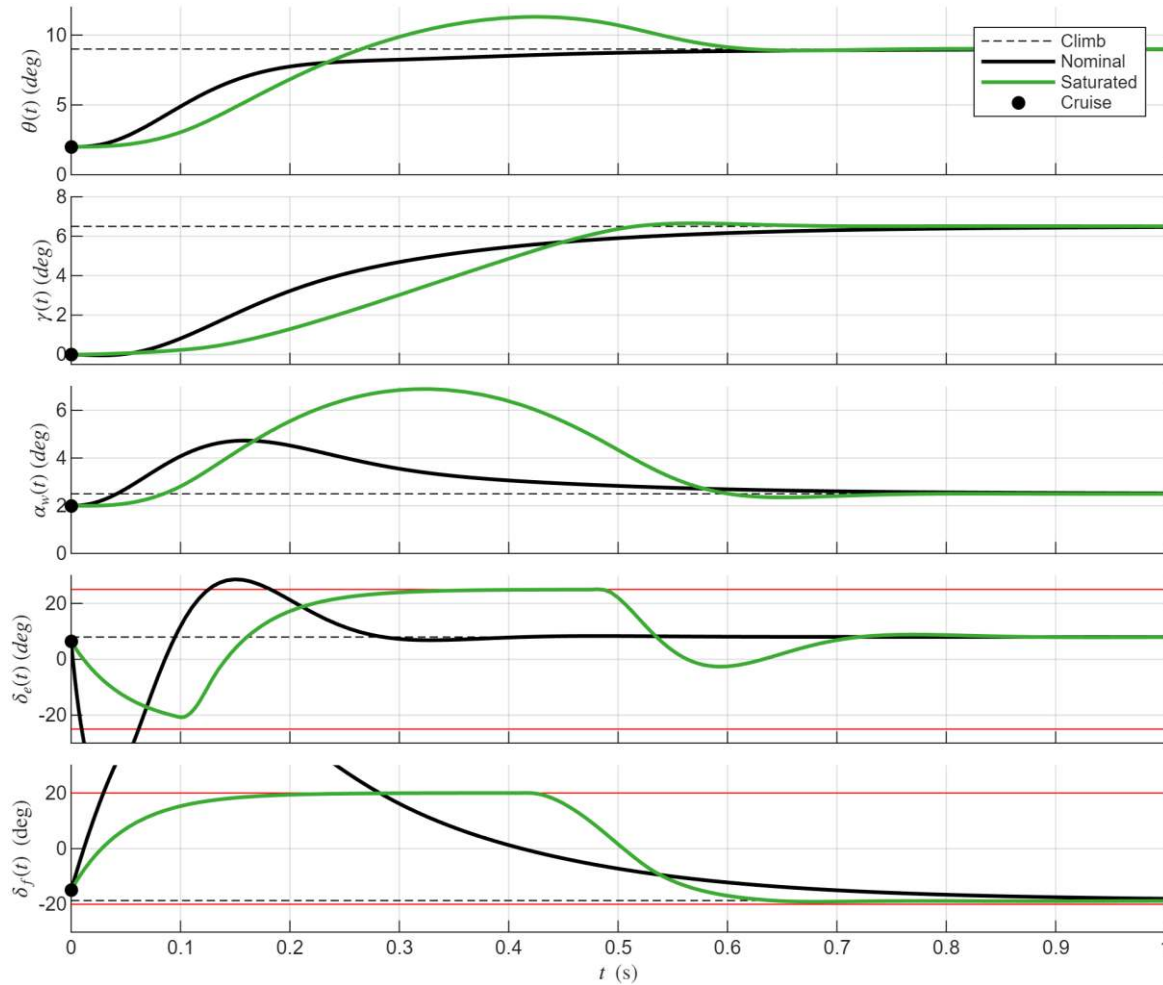
$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$

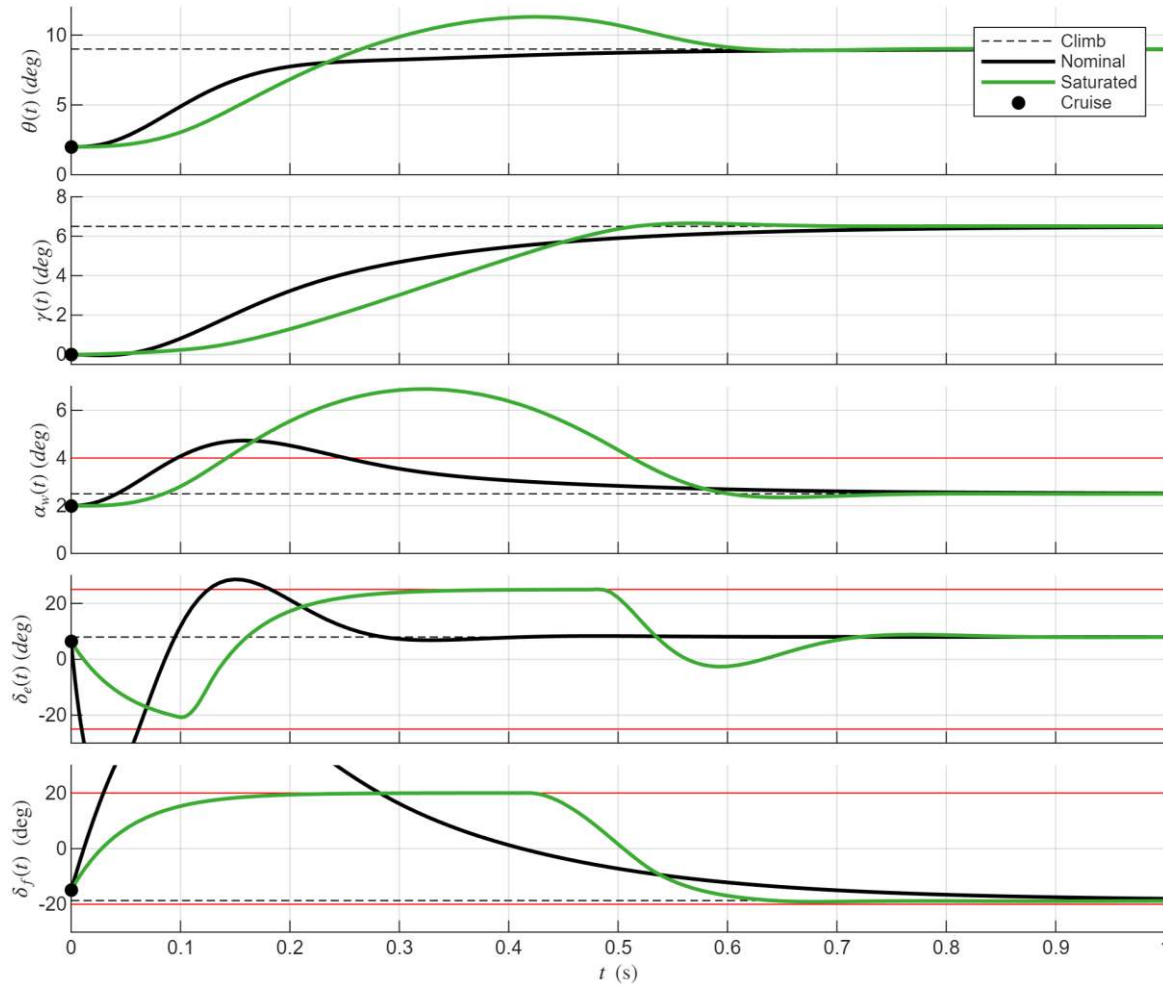


LQR Controller $\kappa(\mathbf{x}) = \bar{\mathbf{u}}(\mathbf{r}) - \mathbf{K}(\mathbf{x} - \bar{\mathbf{x}}(\mathbf{r}))$
Saturated LQR $\mathbf{u} = \text{sat}(\kappa(\mathbf{x}))$

Control surface deflection limits
 $|\delta_e| \leq 25 \text{ deg}, |\delta_f| \leq 20 \text{ deg}$



Example: Safety Filter for F-16



F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$



LQR Controller $\kappa(\mathbf{x}) = \bar{\mathbf{u}}(\mathbf{r}) - \mathbf{K}(\mathbf{x} - \bar{\mathbf{x}}(\mathbf{r}))$
Saturated LQR $\mathbf{u} = \text{sat}(\kappa(\mathbf{x}))$

Pitch pointing constraint

$$|\alpha_w| \leq 25 \text{ deg}$$

Control surface deflection limits

$$|\delta_e| \leq 25 \text{ deg}, |\delta_f| \leq 20 \text{ deg}$$

Example: Safety Filter for F-16

F-16 Pitch Dynamics

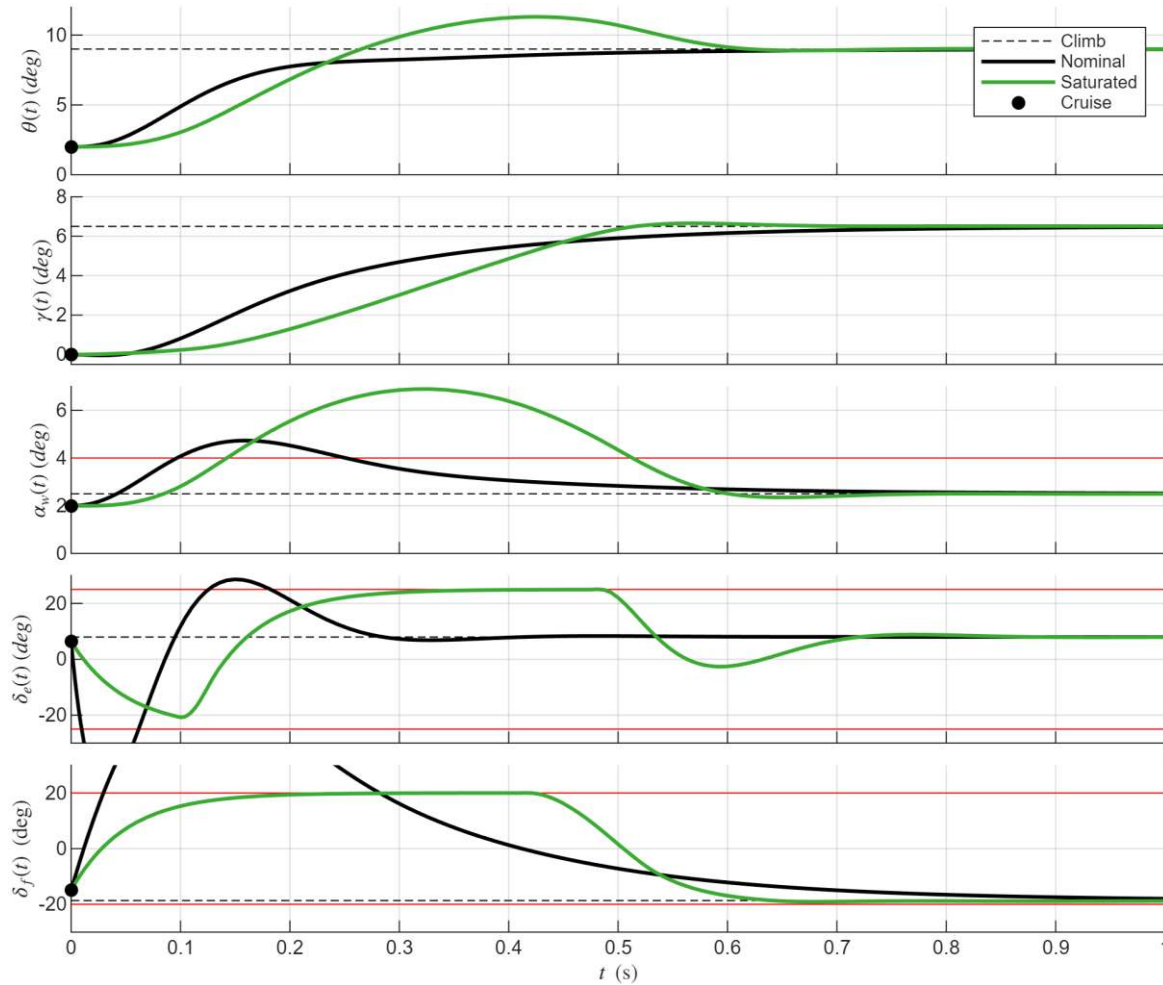
$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$



LQR Controller $\kappa(\mathbf{x}) = \bar{\mathbf{u}}(\mathbf{r}) - \mathbf{K}(\mathbf{x} - \bar{\mathbf{x}}(\mathbf{r}))$
Saturated LQR $\mathbf{u} = \text{sat}(\kappa(\mathbf{x}))$

*Saturation is not good enough.
Let's try CBFs*



Example: Safety Filter for F-16

Naïve CBF Design (elevator)

F-16 Pitch Dynamics

$$x = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$u = [\delta_{e,c} \quad \delta_{f,c}]$$



$$25 \text{ deg} \geq \delta_e$$



This procedure yields *candidate CBFs*.
They provide **no safety guarantees**.

Example: Safety Filter for F-16

Naïve CBF Design (elevator)

Elevator constraint: $h(x) = 25 - \delta_e \geq 0$

F-16 Pitch Dynamics

$$x = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$u = [\delta_{e,c} \quad \delta_{f,c}]$$



$$25 \text{ deg} \geq \delta_e$$




This procedure yields *candidate CBFs*.
They provide **no safety guarantees**.

Example: Safety Filter for F-16

Naïve CBF Design (elevator)

Elevator constraint: $h(\mathbf{x}) = 25 - \delta_e \geq 0$

Differentiate: $\frac{\partial h}{\partial \mathbf{x}} = [0 \quad 0 \quad 0 \quad -1 \quad 0]$

F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$



This procedure yields *candidate CBFs*.
They provide **no safety guarantees**.

Example: Safety Filter for F-16

Naïve CBF Design (elevator)

Elevator constraint: $h(x) = 25 - \delta_e \geq 0$

Differentiate: $\frac{\partial h}{\partial x} = [0 \ 0 \ 0 \ -1 \ 0]$

Compute: $\dot{h}(x, u) = L_f h(x) + L_g h(x)u$

$$L_f h(x) = \frac{\partial h}{\partial x} Ax$$

$$L_g h(x)u = \frac{\partial h}{\partial x} Bu$$



This procedure yields *candidate CBFs*.
They provide **no safety guarantees**.

F-16 Pitch Dynamics

$$x = [\theta \ \dot{\theta} \ \alpha_w \ \delta_e \ \delta_f]$$

$$u = [\delta_{e,c} \ \delta_{f,c}]$$

$$f(x) = Ax \quad g(x) = B$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & -0.87 & 43.22 & -17.25 & -1.58 \\ 0 & 0.99 & -1.34 & -0.17 & -0.25 \\ 0 & 0 & 0 & -20 & 0 \\ 0 & 0 & 0 & 0 & -20 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 20 & 0 \\ 0 & 20 \end{bmatrix}$$



Example: Safety Filter for F-16

Naïve CBF Design (elevator)

Elevator constraint: $h(x) = 25 - \delta_e \geq 0$

Differentiate: $\frac{\partial h}{\partial x} = [0 \ 0 \ 0 \ -1 \ 0]$

Compute: $\dot{h}(x, u) = L_f h(x) + L_g h(x)u$

$$L_f h(x) = \frac{\partial h}{\partial x} A x = 20\delta_e$$

$$L_g h(x)u = \frac{\partial h}{\partial x} B u = -20\delta_{e,c}$$



This procedure yields *candidate CBFs*.
They provide **no safety guarantees**.

F-16 Pitch Dynamics

$$x = [\theta \ \dot{\theta} \ \alpha_w \ \delta_e \ \delta_f]$$

$$u = [\delta_{e,c} \ \delta_{f,c}]$$

$$f(x) = Ax \quad g(x) = B$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & -0.87 & 43.22 & -17.25 & -1.58 \\ 0 & 0.99 & -1.34 & -0.17 & -0.25 \\ 0 & 0 & 0 & -20 & 0 \\ 0 & 0 & 0 & 0 & -20 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 20 & 0 \\ 0 & 20 \end{bmatrix}$$



Example: Safety Filter for F-16

Naïve CBF Design (elevator)

Elevator constraint: $h(\mathbf{x}) = 25 - \delta_e \geq 0$

Differentiate: $\frac{\partial h}{\partial \mathbf{x}} = [0 \quad 0 \quad 0 \quad -1 \quad 0]$

Compute: $\dot{h}(\mathbf{x}, \mathbf{u}) = 20(\delta_e - \delta_{e,c})$

F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$

$$f(\mathbf{x}) = A\mathbf{x} \quad g(\mathbf{x}) = B$$



$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & -0.87 & 43.22 & -17.25 & -1.58 \\ 0 & 0.99 & -1.34 & -0.17 & -0.25 \\ 0 & 0 & 0 & -20 & 0 \\ 0 & 0 & 0 & 0 & -20 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 20 & 0 \\ 0 & 20 \end{bmatrix}$$



This procedure yields *candidate CBFs*.
They provide **no safety guarantees**.

Example: Safety Filter for F-16

Naïve CBF Design (elevator)

Elevator constraint: $h(\mathbf{x}) = 25 - \delta_e \geq 0$

Differentiate: $\frac{\partial h}{\partial \mathbf{x}} = [0 \quad 0 \quad 0 \quad -1 \quad 0]$

Compute: $\dot{h}(\mathbf{x}, \mathbf{u}) = 20(\delta_e - \delta_{e,c})$

Enforce: $\dot{h}(\mathbf{x}, \mathbf{u}) \geq -\alpha(h(\mathbf{x}))$



This procedure yields *candidate CBFs*.
They provide **no safety guarantees**.

F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$



$$\begin{aligned} \min_{\mathbf{u} \in \mathbb{R}^2} \quad & \|\mathbf{u} - \kappa(\mathbf{x})\|^2 \\ \text{s.t.} \quad & \underbrace{20(\delta_e - \delta_{e,c})}_{\dot{h}(\mathbf{x}, \mathbf{u})} \geq -\alpha \underbrace{(25 - \delta_e)}_{h(\mathbf{x})} \end{aligned}$$

Example: Safety Filter for F-16

Naïve CBF Design (elevator)

Elevator constraint: $h(x) = 25 - \delta_e \geq 0$

Differentiate: $\frac{\partial h}{\partial x} = [0 \ 0 \ 0 \ -1 \ 0]$

Compute: $\dot{h}(x, u) = 20(\delta_e - \delta_{e,c})$

Enforce: $\dot{h}(x, u) \geq -\alpha(h(x))$

Tune: $\alpha > 0$

Safety ----- Performance
 $\alpha = 0.1$ $\alpha = 100$



This procedure yields *candidate CBFs*.
They provide **no safety guarantees**.

F-16 Pitch Dynamics

$$x = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$u = [\delta_{e,c} \quad \delta_{f,c}]$$



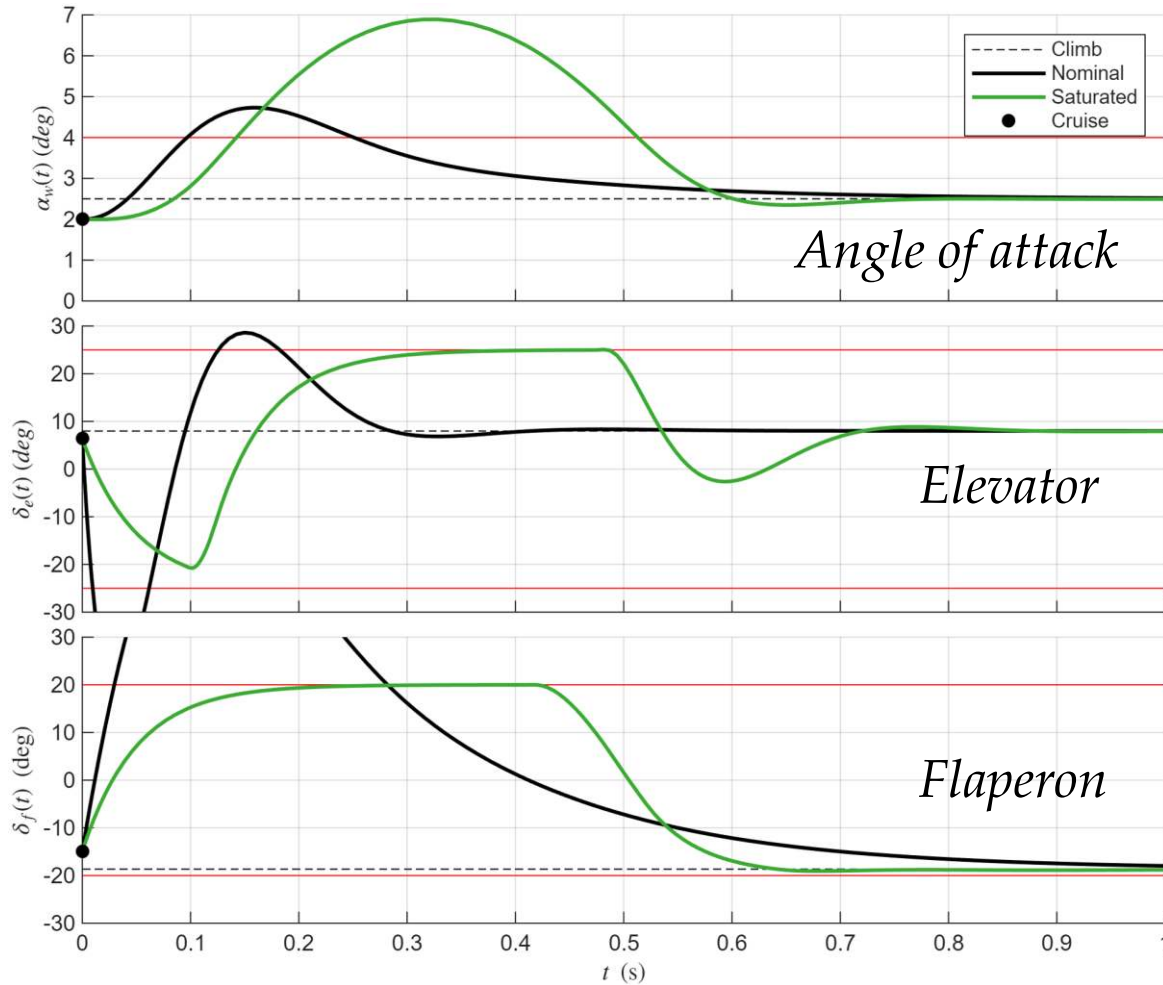
$$\begin{aligned} \min_{u \in \mathbb{R}^2} \quad & \|u - \kappa(x)\|^2 \\ \text{s.t.} \quad & \underbrace{20(\delta_e - \delta_{e,c})}_{\dot{h}(x, u)} \geq -\alpha \underbrace{(25 - \delta_e)}_{h(x)} \end{aligned}$$

Example: Safety Filter for F-16

F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$



LQR Controller $\kappa(\mathbf{x}) = \bar{\mathbf{u}}(\mathbf{r}) - \mathbf{K}(\mathbf{x} - \bar{\mathbf{x}}(\mathbf{r}))$

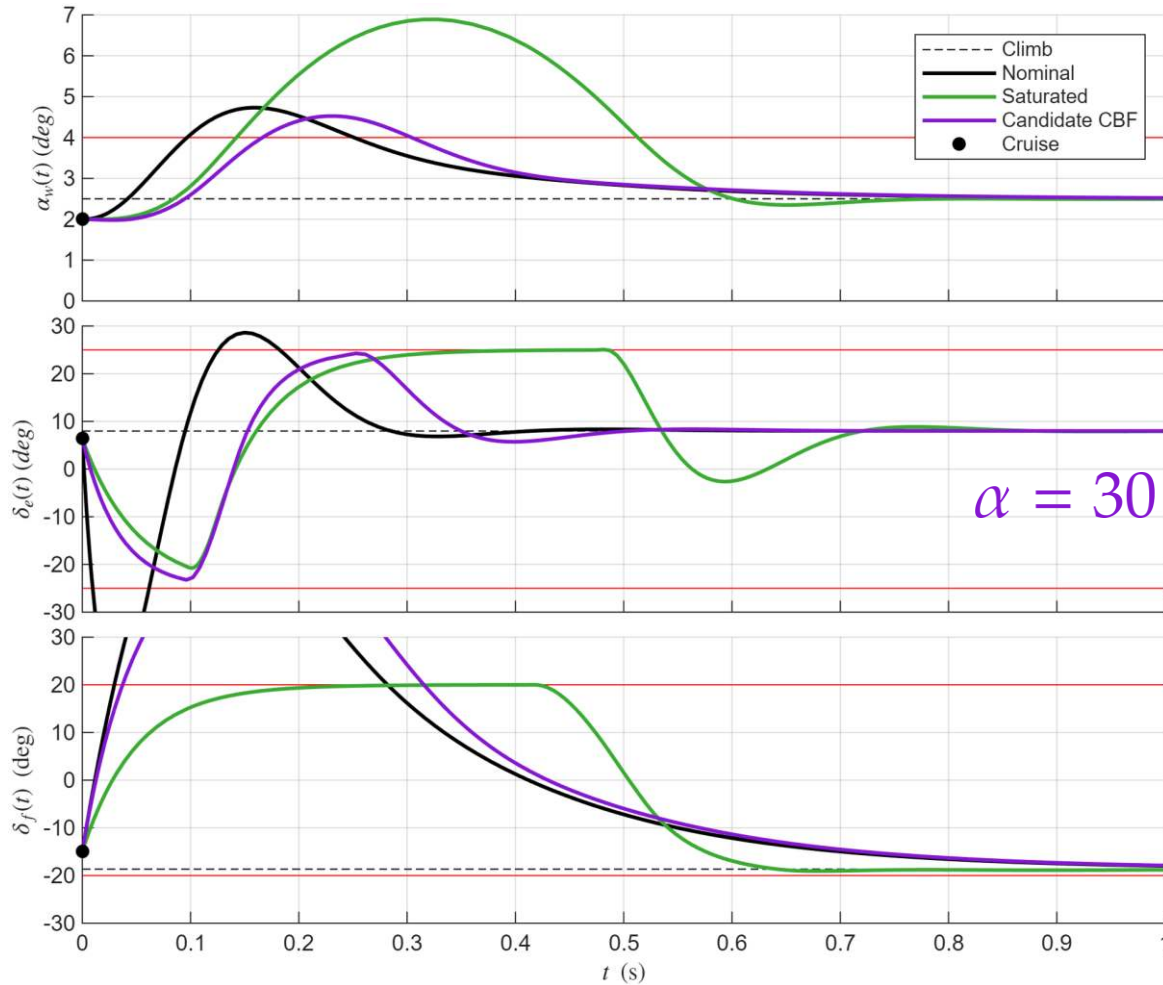
Saturated LQR $\mathbf{u} = \text{sat}(\kappa(\mathbf{x}))$

Example: Safety Filter for F-16

F-16 Pitch Dynamics

$$\mathbf{x} = \begin{bmatrix} \theta & \dot{\theta} & \alpha_w & \delta_e & \delta_f \end{bmatrix}$$

$$\mathbf{u} = \begin{bmatrix} \delta_{e,c} & \delta_{f,c} \end{bmatrix}$$



$$\min_{\mathbf{u} \in \mathbb{R}^2} \|\mathbf{u} - \kappa(\mathbf{x})\|^2$$

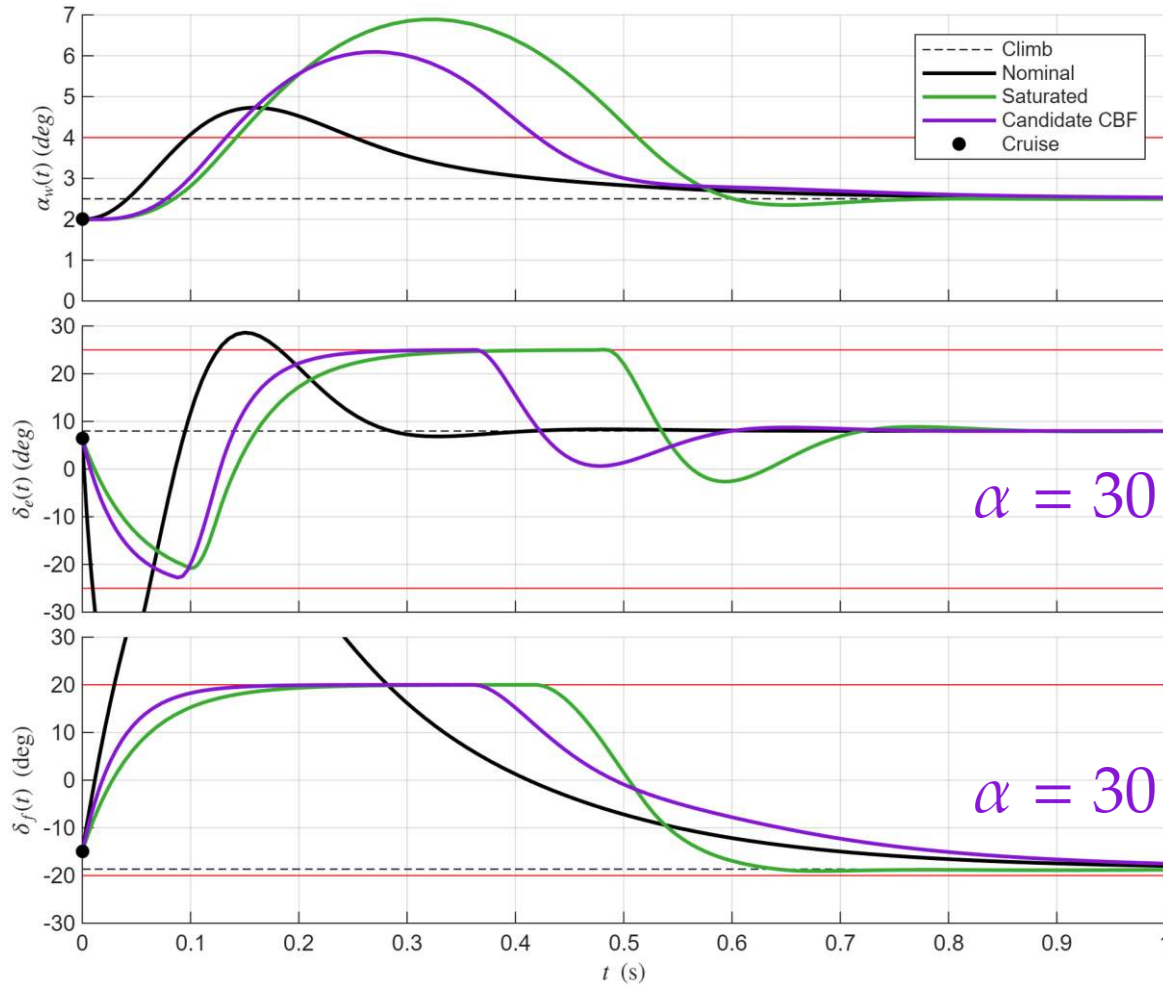
$$\text{s.t. } 20(\delta_e - \delta_{e,c}) \geq -30(25 - \delta_e)$$

Example: Safety Filter for F-16

F-16 Pitch Dynamics

$$\mathbf{x} = \begin{bmatrix} \theta & \dot{\theta} & \alpha_w & \delta_e & \delta_f \end{bmatrix}$$

$$\mathbf{u} = \begin{bmatrix} \delta_{e,c} & \delta_{f,c} \end{bmatrix}$$



$$\min_{\mathbf{u} \in \mathbb{R}^2} \|\mathbf{u} - \kappa(\mathbf{x})\|^2$$

$$\text{s.t.} \quad 20(\delta_e - \delta_{e,c}) \geq -30(25 - \delta_e)$$

$$20(\delta_f - \delta_{f,c}) \geq -30(20 - \delta_f)$$

Example: Safety Filter for F-16

Naïve CBF Design (alpha)

F-16 Pitch Dynamics

$$x = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$u = [\delta_{e,c} \quad \delta_{f,c}]$$



$$4 \text{ deg} \geq \alpha_w$$



This procedure yields *candidate CBFs*.
They provide **no safety guarantees**.

Example: Safety Filter for F-16

Naïve CBF Design (alpha)

Stall constraint: $h(\mathbf{x}) = 4 - \alpha_w \geq 0$



This procedure yields *candidate CBFs*.
They provide **no safety guarantees**.

F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$



$$4 \text{ deg} \geq \alpha_w$$

Example: Safety Filter for F-16

Naïve CBF Design (alpha)

Stall constraint: $h(\mathbf{x}) = 4 - \alpha_w \geq 0$

Differentiate: $\frac{\partial h}{\partial \mathbf{x}} = [0 \quad 0 \quad -1 \quad 0 \quad 0]$



This procedure yields *candidate CBFs*.
They provide **no safety guarantees**.

F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$



Example: Safety Filter for F-16

Naïve CBF Design (alpha)

Stall constraint: $h(x) = 4 - \alpha_w \geq 0$

Differentiate: $\frac{\partial h}{\partial x} = [0 \quad 0 \quad -1 \quad 0 \quad 0]$

Compute: $\dot{h}(x, u) = L_f h(x) + L_g h(x)u$

$$L_f h(x) = \frac{\partial h}{\partial x} A x$$

$$L_g h(x) = \frac{\partial h}{\partial x} B$$



This procedure yields *candidate CBFs*.
They provide **no safety guarantees**.

F-16 Pitch Dynamics

$$x = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$u = [\delta_{e,c} \quad \delta_{f,c}]$$

$$f(x) = Ax \quad g(x) = B$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & -0.87 & 43.22 & -17.25 & -1.58 \\ 0 & 0.99 & -1.34 & -0.17 & -0.25 \\ 0 & 0 & 0 & -20 & 0 \\ 0 & 0 & 0 & 0 & -20 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 20 & 0 \\ 0 & 20 \end{bmatrix}$$



Example: Safety Filter for F-16

Naïve CBF Design (alpha)

Stall constraint: $h(x) = 4 - \alpha_w \geq 0$

Differentiate: $\frac{\partial h}{\partial x} = [0 \quad 0 \quad -1 \quad 0 \quad 0]$

Compute: $\dot{h}(x, u) = L_f h(x) + L_g h(x)u$

$$\begin{aligned} L_f h(x) &= \frac{\partial h}{\partial x} A x \\ &= -\dot{\theta} + 1.3\alpha_w + 0.2\delta_e + 0.3\delta_f \end{aligned}$$

$$L_g h(x) = \frac{\partial h}{\partial x} B$$



This procedure yields *candidate CBFs*.
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F-16 Pitch Dynamics

$$x = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$u = [\delta_{e,c} \quad \delta_{f,c}]$$

$$f(x) = Ax \quad g(x) = B$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & -0.87 & 43.22 & -17.25 & -1.58 \\ 0 & 0.99 & -1.34 & -0.17 & -0.25 \\ 0 & 0 & 0 & -20 & 0 \\ 0 & 0 & 0 & 0 & -20 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 20 & 0 \\ 0 & 20 \end{bmatrix}$$



Example: Safety Filter for F-16

Naïve CBF Design (alpha)

Stall constraint: $h(\mathbf{x}) = 4 - \alpha_w \geq 0$

Differentiate: $\frac{\partial h}{\partial \mathbf{x}} = [0 \quad 0 \quad -1 \quad 0 \quad 0]$

Compute: $\dot{h}(\mathbf{x}, \mathbf{u}) = L_f h(\mathbf{x}) + L_g h(\mathbf{x}) \mathbf{u}$

$$\begin{aligned} L_f h(\mathbf{x}) &= \frac{\partial h}{\partial \mathbf{x}} A \mathbf{x} \\ &= -\dot{\theta} + 1.3\alpha_w + 0.2\delta_e + 0.3\delta_f \end{aligned}$$

$$L_g h(\mathbf{x}) = \frac{\partial h}{\partial \mathbf{x}} B = [0 \quad 0]$$



This procedure yields *candidate CBFs*.
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F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$

$$f(\mathbf{x}) = A\mathbf{x} \quad g(\mathbf{x}) = B$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & -0.87 & 43.22 & -17.25 & -1.58 \\ 0 & 0.99 & -1.34 & -0.17 & -0.25 \\ 0 & 0 & 0 & -20 & 0 \\ 0 & 0 & 0 & 0 & -20 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 20 & 0 \\ 0 & 20 \end{bmatrix}$$



Example: Safety Filter for F-16

Naïve CBF Design (alpha)

Stall constraint: $h(x) = 4 - \alpha_w \geq 0$

Differentiate: $\frac{\partial h}{\partial x} = [0 \quad 0 \quad -1 \quad 0 \quad 0]$

Compute: $\dot{h}(x, u) = L_f h(x) + L_g h(x)u$

$$\begin{aligned} L_f h(x) &= \frac{\partial h}{\partial x} A x \\ &= -\dot{\theta} + 1.3\alpha_w + 0.2\delta_e + 0.3\delta_f \end{aligned}$$

$$L_g h(x) = \frac{\partial h}{\partial x} B = [0 \quad 0]$$



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F-16 Pitch Dynamics

$$x = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$u = [\delta_{e,c} \quad \delta_{f,c}]$$

$$f(x) = Ax \quad g(x) = B$$



Relative degree > 1

Example: Safety Filter for F-16

Naïve CBF Design (alpha)

Stall constraint: $h(x) = 4 - \alpha_w \geq 0$

Differentiate: $\frac{\partial h}{\partial x} = [0 \quad 0 \quad -1 \quad 0 \quad 0]$

Compute: $\dot{h}(x, u) = L_f h(x) + \cancel{L_g h(x)} u$

$$\begin{aligned} L_f h(x) &= \frac{\partial h}{\partial x} A x \\ &= -\dot{\theta} + 1.3\alpha_w + 0.2\delta_e + 0.3\delta_f \end{aligned}$$

$$L_g h(x) = \frac{\partial h}{\partial x} B = [0 \quad 0]$$

Lost Control Authority



This procedure yields *candidate CBFs*.
They provide **no safety guarantees**.

F-16 Pitch Dynamics

$$x = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$u = [\delta_{e,c} \quad \delta_{f,c}]$$

$$f(x) = Ax \quad g(x) = B$$



Example: Safety Filter for F-16

Naïve CBF Design (alpha)

Stall constraint: $h(\mathbf{x}) = 4 - \alpha_w \geq 0$

Differentiate: $\frac{\partial h}{\partial \mathbf{x}} = [0 \quad 0 \quad -1 \quad 0 \quad 0]$

Compute: $\dot{h}(\mathbf{x}, \mathbf{u}) = L_f h(\mathbf{x})$

Enforce: $L_f h(\mathbf{x}) \geq -\alpha(h(\mathbf{x}))$

F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$



Lost Control Authority



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Example: Safety Filter for F-16

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Stall constraint: $h(\mathbf{x}) = 4 - \alpha_w \geq 0$

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F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$



Example: Safety Filter for F-16

Naïve CBF Design (alpha)

Stall constraint: $h(\mathbf{x}) = 4 - \alpha_w \geq 0$

Enforce: $L_f h(\mathbf{x}) \geq -\alpha(h(\mathbf{x}))$

Treat as new constraint...

Define: $h_1(\mathbf{x}) = L_f h(\mathbf{x}) + \alpha(h(\mathbf{x})) \geq 0$



This procedure yields *candidate CBFs*.
They provide **no safety guarantees**.

F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$



Example: Safety Filter for F-16

Naïve CBF Design (alpha)

Stall constraint: $h(\mathbf{x}) = 4 - \alpha_w \geq 0$

Enforce: $L_f h(\mathbf{x}) \geq -\alpha(h(\mathbf{x}))$

Treat as new constraint...

Define: $h_1(\mathbf{x}) = L_f h(\mathbf{x}) + \alpha(h(\mathbf{x})) \geq 0$

Differentiate: $\frac{\partial h_1}{\partial \mathbf{x}} = \frac{\partial h}{\partial \mathbf{x}}(A + \alpha I_n)$



This procedure yields *candidate CBFs*.
They provide **no safety guarantees**.

F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$



$$\frac{\partial h}{\partial \mathbf{x}} = [0 \quad 0 \quad -1 \quad 0 \quad 0]$$

Example: Safety Filter for F-16

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Stall constraint: $h(\mathbf{x}) = 4 - \alpha_w \geq 0$

Enforce: $L_f h(\mathbf{x}) \geq -\alpha(h(\mathbf{x}))$

Treat as new constraint...

Define: $h_1(\mathbf{x}) = L_f h(\mathbf{x}) + \alpha(h(\mathbf{x})) \geq 0$

Differentiate: $\frac{\partial h_1}{\partial \mathbf{x}} = \frac{\partial h}{\partial \mathbf{x}}(A + \alpha I_n)$

Compute: $\dot{h}_1(\mathbf{x}, \mathbf{u}) = L_f h_1(\mathbf{x}) + L_g h_1(\mathbf{x}) \mathbf{u}$



This procedure yields **candidate CBFs**.
They provide **no safety guarantees**.

F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$

$$f(\mathbf{x}) = A\mathbf{x} \quad g(\mathbf{x}) = B$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & -0.87 & 43.22 & -17.25 & -1.58 \\ 0 & 0.99 & -1.34 & -0.17 & -0.25 \\ 0 & 0 & 0 & -20 & 0 \\ 0 & 0 & 0 & 0 & -20 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 20 & 0 \\ 0 & 20 \end{bmatrix}$$

$$\frac{\partial h}{\partial \mathbf{x}} = [0 \quad 0 \quad -1 \quad 0 \quad 0]$$



Example: Safety Filter for F-16

Naïve CBF Design (alpha)

Stall constraint: $h(x) = 4 - \alpha_w \geq 0$

Enforce: $L_f h(x) \geq -\alpha(h(x))$

Treat as new constraint...

Define: $h_1(x) = L_f h(x) + \alpha(h(x)) \geq 0$

Differentiate: $\frac{\partial h_1}{\partial x} = \frac{\partial h}{\partial x}(A + \alpha I_n)$

Compute: $\dot{h}_1(x, u) = L_f h_1(x) + L_g h_1(x)u$

$$L_g h_1(x) = \frac{\partial h_1}{\partial x} B$$



This procedure yields **candidate CBFs**.
They provide **no safety guarantees**.

F-16 Pitch Dynamics

$$x = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$u = [\delta_{e,c} \quad \delta_{f,c}]$$

$$f(x) = Ax \quad g(x) = B$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & -0.87 & 43.22 & -17.25 & -1.58 \\ 0 & 0.99 & -1.34 & -0.17 & -0.25 \\ 0 & 0 & 0 & -20 & 0 \\ 0 & 0 & 0 & 0 & -20 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 20 & 0 \\ 0 & 20 \end{bmatrix}$$

$$\frac{\partial h}{\partial x} = [0 \quad 0 \quad -1 \quad 0 \quad 0]$$



Example: Safety Filter for F-16

Naïve CBF Design (alpha)

Stall constraint: $h(x) = 4 - \alpha_w \geq 0$

Enforce: $L_f h(x) \geq -\alpha(h(x))$

Treat as new constraint...

Define: $h_1(x) = L_f h(x) + \alpha(h(x)) \geq 0$

Differentiate: $\frac{\partial h_1}{\partial x} = \frac{\partial h}{\partial x}(A + \alpha I_n)$

Compute: $\dot{h}_1(x, u) = L_f h_1(x) + L_g h_1(x)u$

$$L_g h_1(x) = \frac{\partial h_1}{\partial x} B = \frac{\partial h}{\partial x}(A + \alpha I_n)B$$



This procedure yields **candidate CBFs**.
They provide **no safety guarantees**.

F-16 Pitch Dynamics

$$x = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$u = [\delta_{e,c} \quad \delta_{f,c}]$$

$$f(x) = Ax \quad g(x) = B$$



$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & -0.87 & 43.22 & -17.25 & -1.58 \\ 0 & 0.99 & -1.34 & -0.17 & -0.25 \\ 0 & 0 & 0 & -20 & 0 \\ 0 & 0 & 0 & 0 & -20 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 20 & 0 \\ 0 & 20 \end{bmatrix}$$

$$\frac{\partial h}{\partial x} = [0 \quad 0 \quad -1 \quad 0 \quad 0]$$

Example: Safety Filter for F-16

Naïve CBF Design (alpha)

Stall constraint: $h(x) = 4 - \alpha_w \geq 0$

Enforce: $L_f h(x) \geq -\alpha(h(x))$

Treat as new constraint...

Define: $h_1(x) = L_f h(x) + \alpha(h(x)) \geq 0$

Differentiate: $\frac{\partial h_1}{\partial x} = \frac{\partial h}{\partial x}(A + \alpha I_n)$

Compute: $\dot{h}_1(x, u) = L_f h_1(x) + L_g h_1(x)u$

$$L_g h_1(x) = \frac{\partial h_1}{\partial x} B = \frac{\partial h}{\partial x} AB + \alpha \frac{\partial h}{\partial x} B$$



This procedure yields **candidate CBFs**.
They provide **no safety guarantees**.

F-16 Pitch Dynamics

$$x = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$u = [\delta_{e,c} \quad \delta_{f,c}]$$

$$f(x) = Ax \quad g(x) = B$$



$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & -0.87 & 43.22 & -17.25 & -1.58 \\ 0 & 0.99 & -1.34 & -0.17 & -0.25 \\ 0 & 0 & 0 & -20 & 0 \\ 0 & 0 & 0 & 0 & -20 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 20 & 0 \\ 0 & 20 \end{bmatrix}$$

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$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & -0.87 & 43.22 & -17.25 & -1.58 \\ 0 & 0.99 & -1.34 & -0.17 & -0.25 \\ 0 & 0 & 0 & -20 & 0 \\ 0 & 0 & 0 & 0 & -20 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 20 & 0 \\ 0 & 20 \end{bmatrix}$$

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Example: Safety Filter for F-16

Naïve CBF Design (alpha)

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Differentiate: $\frac{\partial h_1}{\partial \mathbf{x}} = \frac{\partial h}{\partial \mathbf{x}}(A + \alpha I_n)$

Compute: $\dot{h}_1(\mathbf{x}, \mathbf{u}) = L_f h_1(\mathbf{x}) + L_g h_1(\mathbf{x}) \mathbf{u}$

$$L_g h_1(\mathbf{x}) = \frac{\partial h_1}{\partial \mathbf{x}} B = \begin{bmatrix} 3.4 & 5.0 \end{bmatrix}$$



This procedure yields *candidate CBFs*.
They provide **no safety guarantees**.

F-16 Pitch Dynamics

$$\mathbf{x} = \begin{bmatrix} \theta & \dot{\theta} & \alpha_w & \delta_e & \delta_f \end{bmatrix}$$

$$\mathbf{u} = \begin{bmatrix} \delta_{e,c} & \delta_{f,c} \end{bmatrix}$$



Recovered Control Authority

Example: Safety Filter for F-16

Naïve CBF Design (alpha)

Stall constraint: $h(\mathbf{x}) = 4 - \alpha_w \geq 0$

Enforce: $L_f h(\mathbf{x}) \geq -\alpha(h(\mathbf{x}))$

Treat as new constraint...

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Compute: $\dot{h}_1(\mathbf{x}, \mathbf{u}) = L_f h_1(\mathbf{x}) + L_g h_1(\mathbf{x}) \mathbf{u}$

Enforce: $\dot{h}_1(\mathbf{x}, \mathbf{u}) \geq -\alpha_1(h_1(\mathbf{x}))$



This procedure yields **candidate CBFs**.
They provide **no safety guarantees**.

F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$



$$\begin{aligned} \min_{\mathbf{u} \in \mathbb{R}^2} \quad & \|\mathbf{u} - \kappa(\mathbf{x})\|^2 \\ \text{s.t.} \quad & \dot{h}_1(\mathbf{x}, \mathbf{u}) \geq -\alpha_1(h_1(\mathbf{x})) \end{aligned}$$

Example: Safety Filter for F-16

Naïve CBF Design (alpha)

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F-16 Pitch Dynamics

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$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$



$$\begin{aligned} \min_{\mathbf{u} \in \mathbb{R}^2} \quad & \|\mathbf{u} - \kappa(\mathbf{x})\|^2 \\ \text{s.t.} \quad & A_{QP} \mathbf{u} \leq b_{QP}(\mathbf{x}) \end{aligned}$$

Example: Safety Filter for F-16

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$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$



$$\min_{\mathbf{u} \in \mathbb{R}^2} \|\mathbf{u} - \kappa(\mathbf{x})\|^2$$

$$\text{s.t. } A_{QP} \mathbf{u} \leq b_{QP}(\mathbf{x})$$

$$A_{QP} = -L_g h_1(\mathbf{x}) = -[3.4 \quad 5.0]$$

Example: Safety Filter for F-16

Naïve CBF Design (alpha)

Stall constraint: $h(\mathbf{x}) = 4 - \alpha_w \geq 0$

Enforce: $L_f h(\mathbf{x}) \geq -\alpha(h(\mathbf{x}))$

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F-16 Pitch Dynamics

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$$A_{QP} = -L_g h_1(\mathbf{x}) = -[3.4 \quad 5.0]$$

$$b_{QP}(\mathbf{x}) = (\alpha + \alpha_1) \frac{\partial h}{\partial \mathbf{x}} A \mathbf{x} + \alpha \alpha_1 h(\mathbf{x}) + \frac{\partial h}{\partial \mathbf{x}} A^2 \mathbf{x}$$

Example: Safety Filter for F-16

Naïve CBF Design (alpha)

Stall constraint: $h(\mathbf{x}) = 4 - \alpha_w \geq 0$

Enforce: $L_f h(\mathbf{x}) \geq -\alpha(h(\mathbf{x}))$

Treat as new constraint...

Define: $h_1(\mathbf{x}) = L_f h(\mathbf{x}) + \alpha(h(\mathbf{x})) \geq 0$

Differentiate: $\frac{\partial h_1}{\partial \mathbf{x}} = \frac{\partial h}{\partial \mathbf{x}}(A + \alpha I_n)$

Compute: $\dot{h}_1(\mathbf{x}, \mathbf{u}) = L_f h_1(\mathbf{x}) + L_g h_1(\mathbf{x}) \mathbf{u}$

Enforce: $\dot{h}_1(\mathbf{x}, \mathbf{u}) \geq -\alpha_1(h_1(\mathbf{x}))$

Tune: $\alpha, \alpha_1 > 0$



This procedure yields **candidate CBFs**.
They provide **no safety guarantees**.

F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$



$$\min_{\mathbf{u} \in \mathbb{R}^2} \|\mathbf{u} - \kappa(\mathbf{x})\|^2$$

$$\text{s.t. } A_{QP} \mathbf{u} \leq b_{QP}(\mathbf{x})$$

*Roots of polynomial,
eigenvalues (2x2)*

$$A_{QP} = -L_g h_1(\mathbf{x}) = -[3.4 \quad 5.0]$$

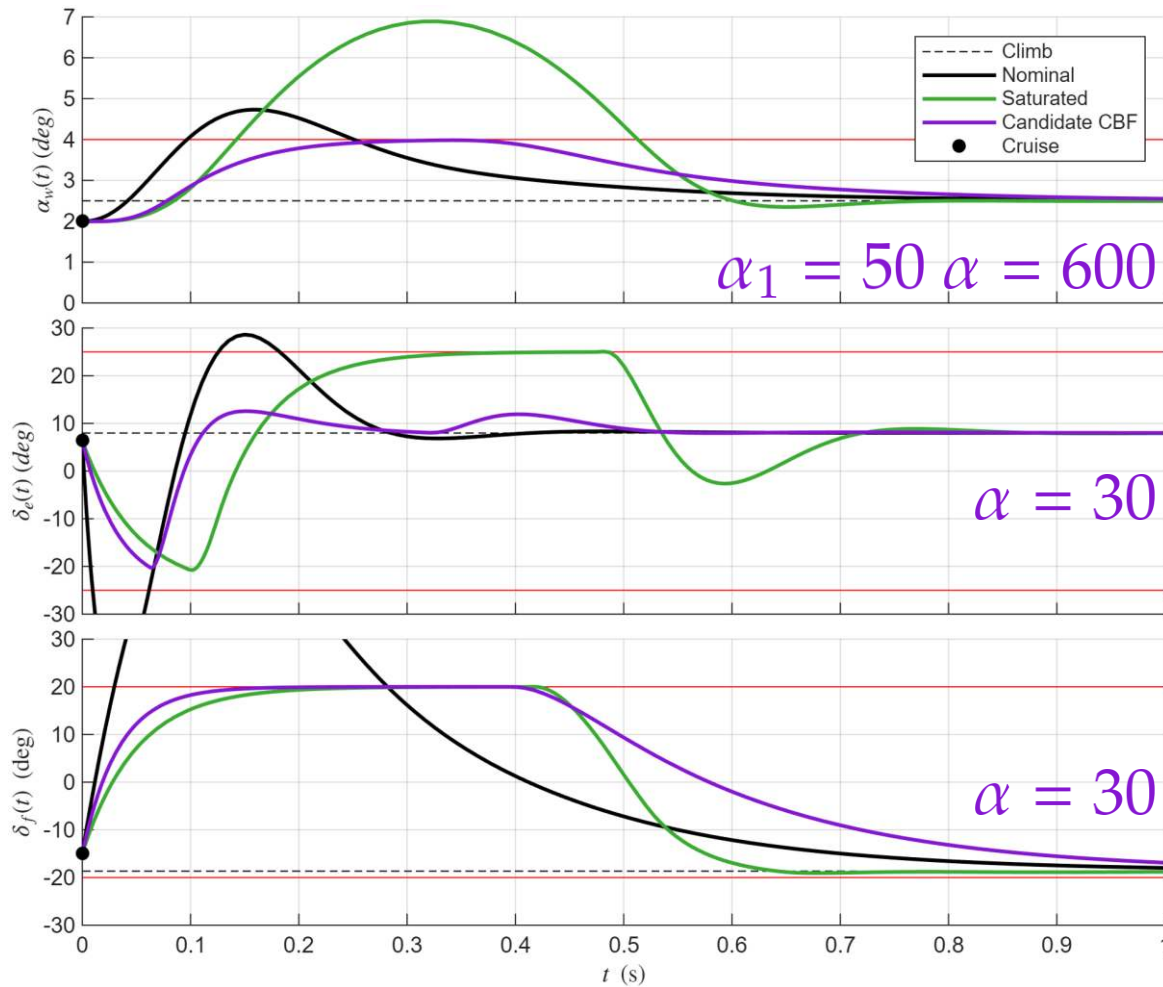
$$b_{QP}(\mathbf{x}) = (\alpha + \alpha_1) \frac{\partial h}{\partial \mathbf{x}} A \mathbf{x} + \alpha \alpha_1 h(\mathbf{x}) + \frac{\partial h}{\partial \mathbf{x}} A^2 \mathbf{x}$$

Example: Safety Filter for F-16

F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$



$$\min_{\mathbf{u} \in \mathbb{R}^2} \|\mathbf{u} - \kappa(\mathbf{x})\|^2$$

$$\text{s.t.} \quad 20(\delta_e - \delta_{e,c}) \geq -30(25 - \delta_e)$$

$$20(\delta_f - \delta_{f,c}) \geq -30(20 - \delta_f)$$

$$\dot{h}_1(\mathbf{x}, \mathbf{u}) \geq -\alpha_1(h_1(\mathbf{x}))$$

mosek

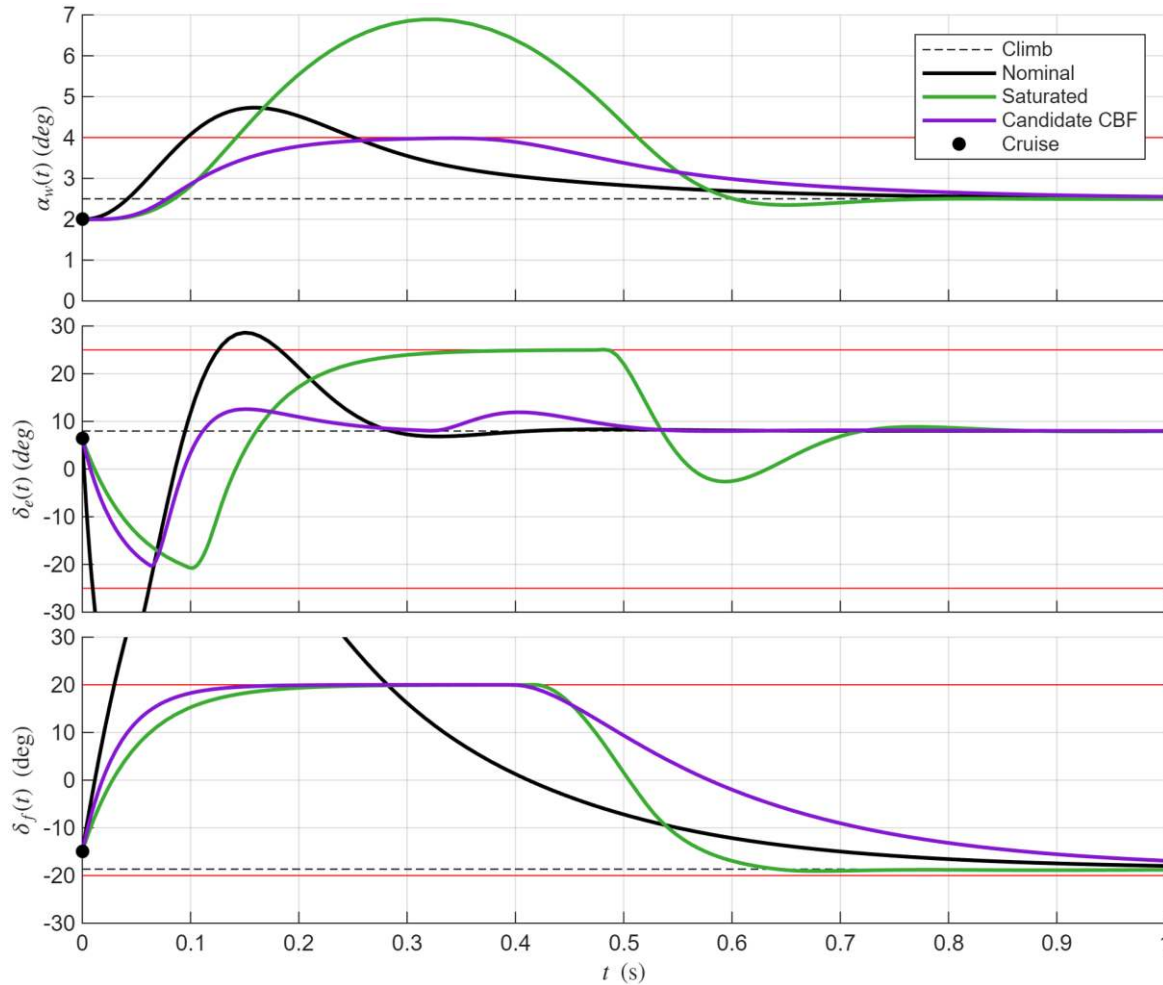
3.5 ms solvetime

Example: Safety Filter for F-16

F-16 Pitch Dynamics

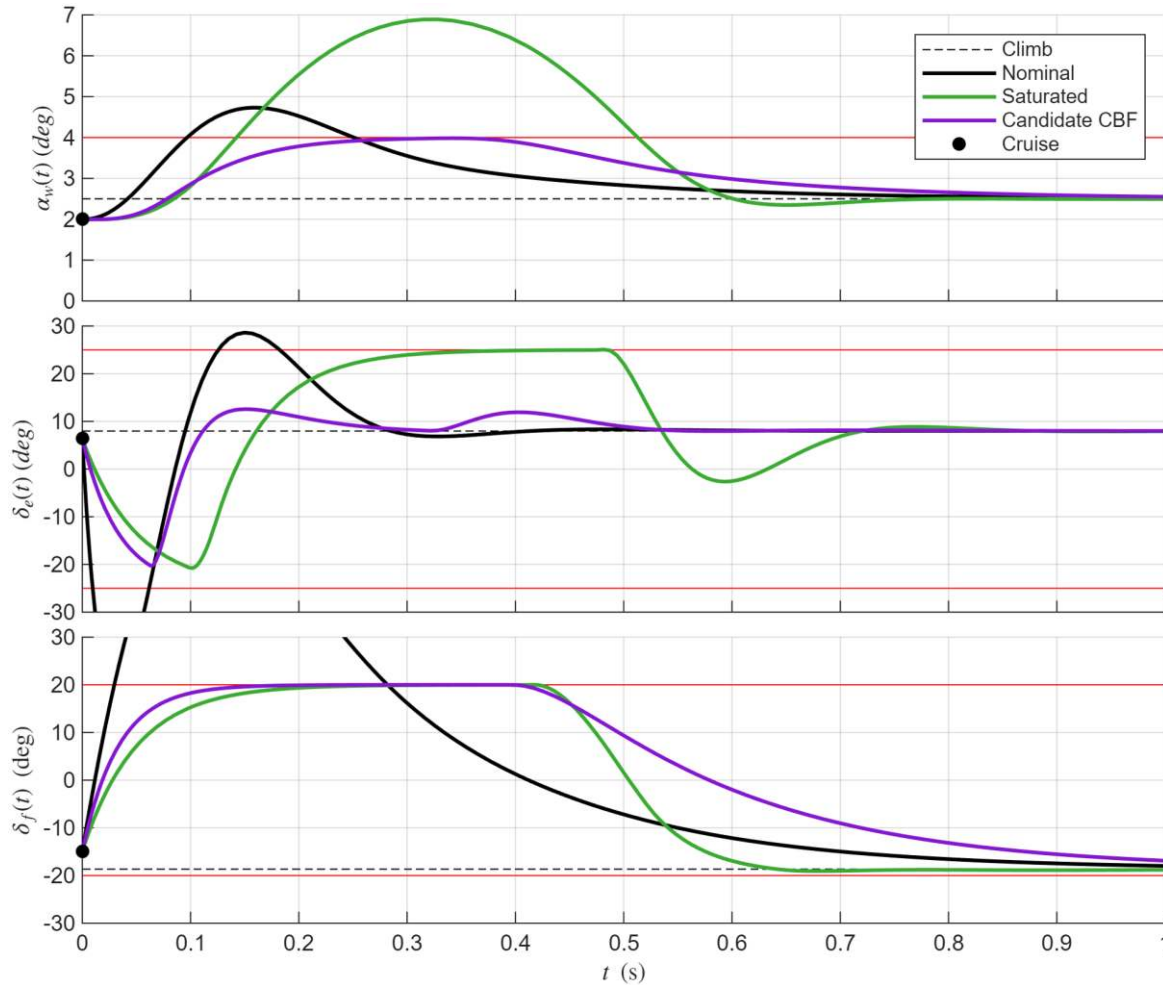
$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

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This procedure yields *candidate CBFs*.
They provide **no safety guarantees**.

Example: Safety Filter for F-16



F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

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Cruise: $\theta = 2, \gamma = 0$



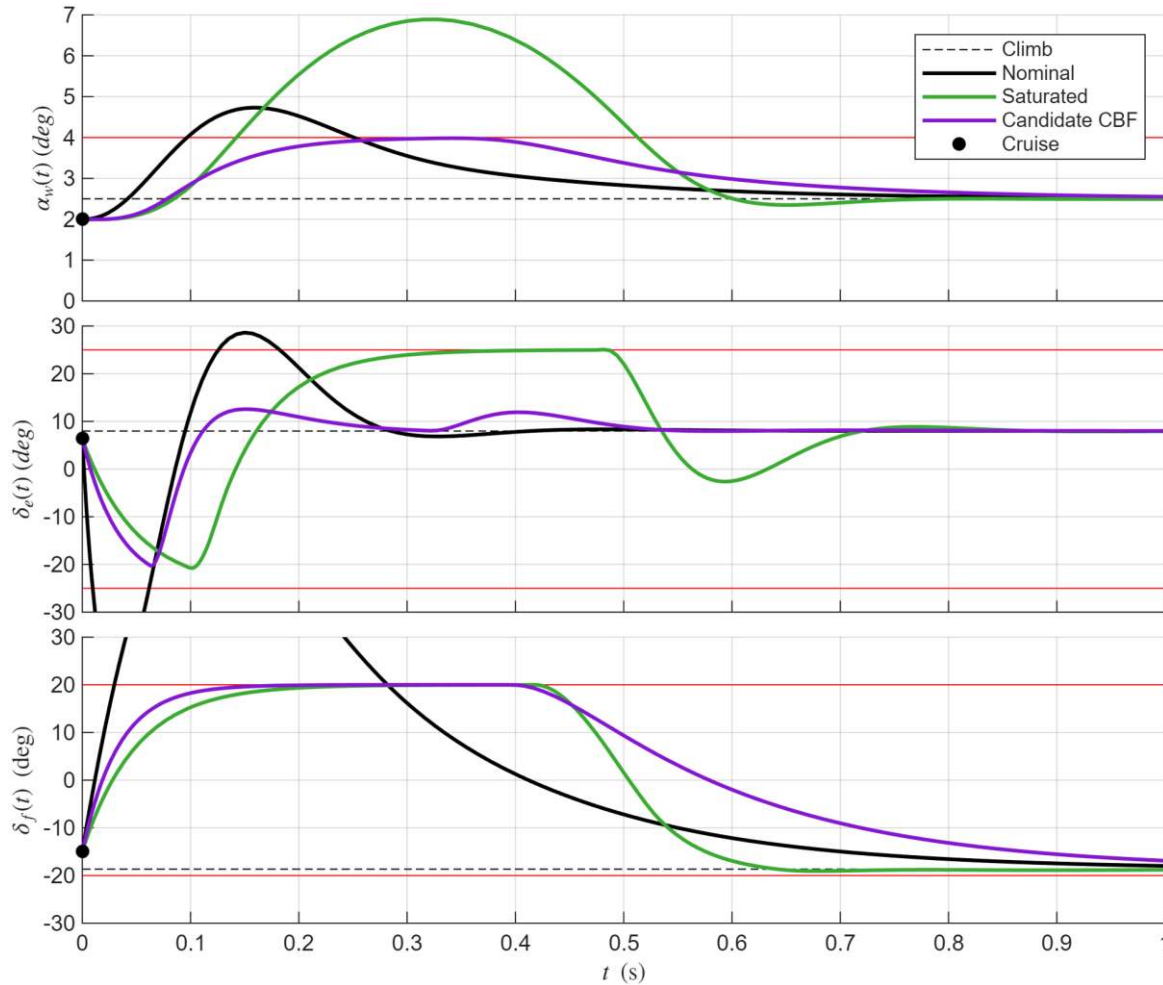
Goal

Climb: $\theta = 9, \gamma = 6.5$



This procedure yields *candidate CBFs*.
They provide **no safety guarantees**.

Example: Safety Filter for F-16



F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

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Cruise: $\theta = 2, \gamma = 0$



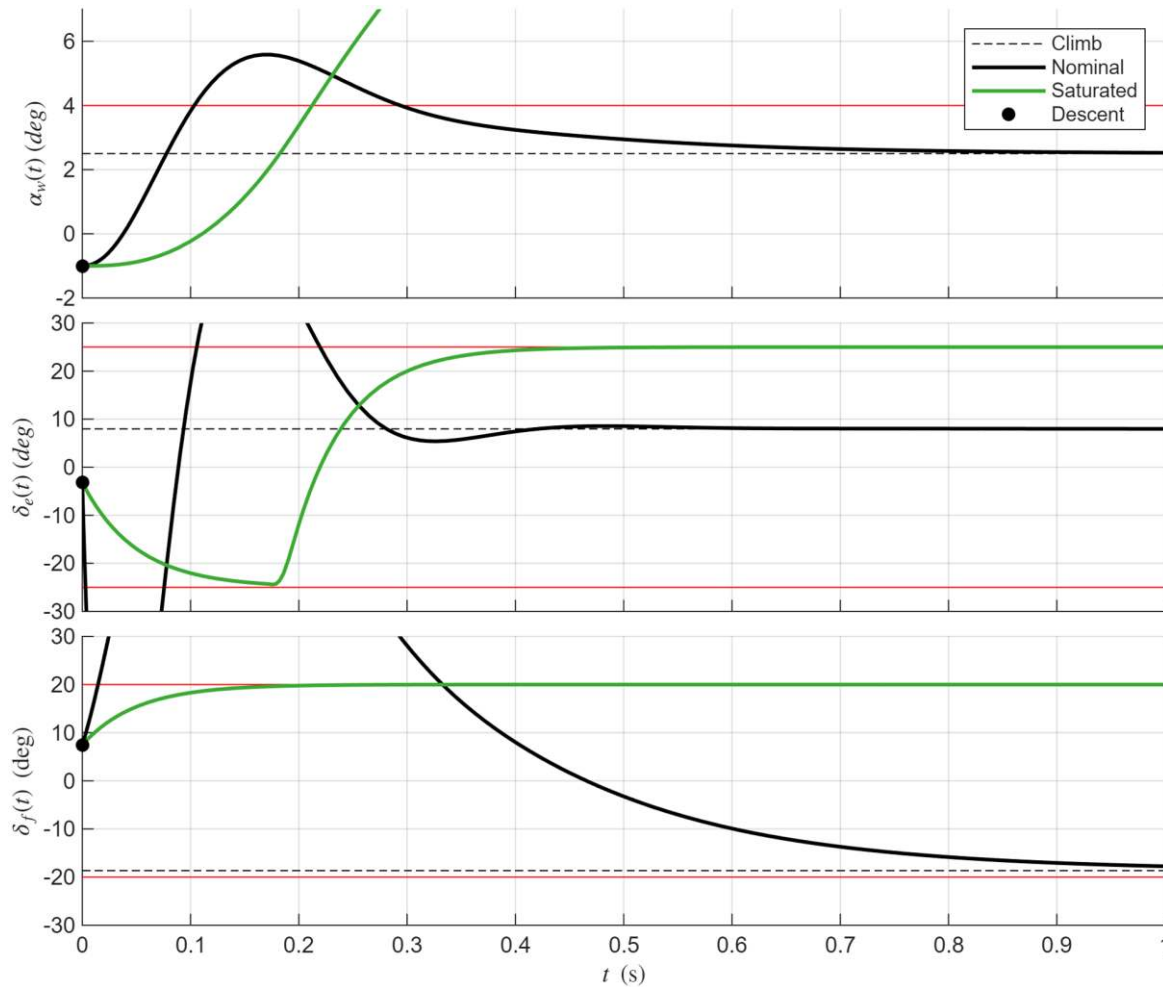
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This procedure yields *candidate CBFs*.
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Example: Safety Filter for F-16



F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$



Descent: $\theta = -3, \gamma = -2$



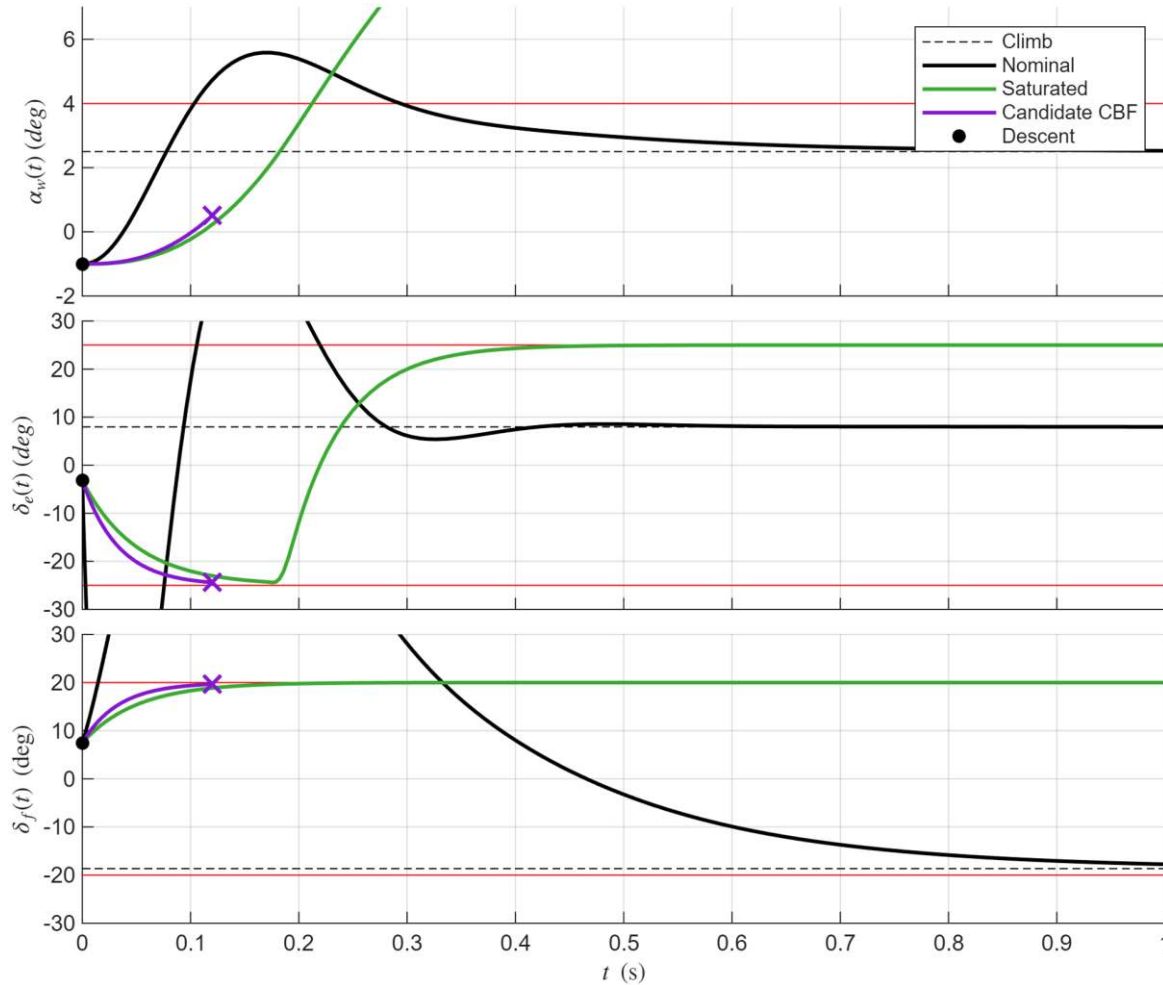
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Climb: $\theta = 9, \gamma = 6.5$



This procedure yields *candidate CBFs*.
They provide **no safety guarantees**.

Example: Safety Filter for F-16



F-16 Pitch Dynamics

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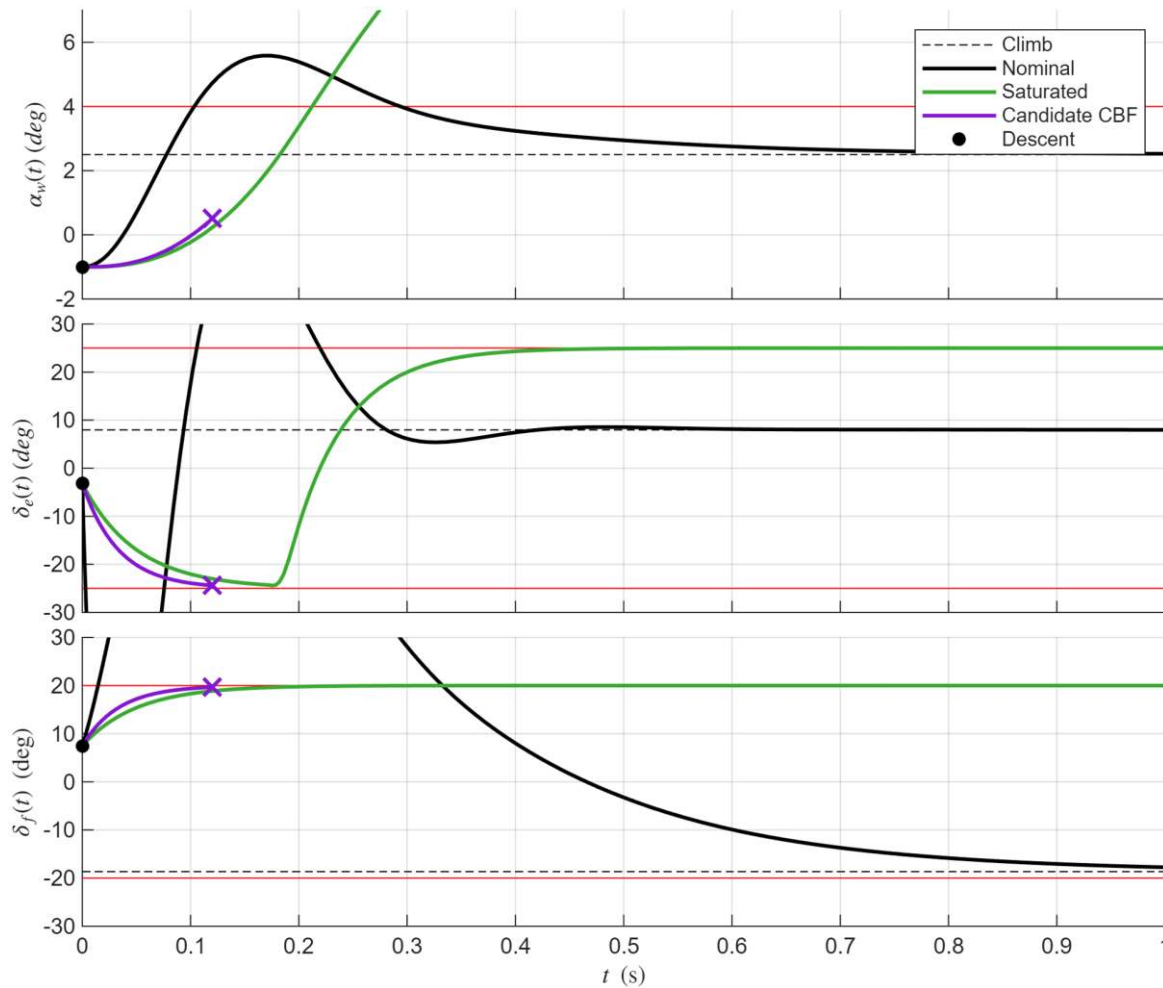
Goal

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This procedure yields *candidate CBFs*.
They provide **no safety guarantees**.

Example: Safety Filter for F-16



F-16 Pitch Dynamics

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$$\min_{\mathbf{u} \in \mathbb{R}^2} \|\mathbf{u} - \kappa(\mathbf{x})\|^2$$

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$$20(\delta_f - \delta_{f,c}) \geq -30(20 - \delta_f)$$

$$\dot{h}_1(\mathbf{x}, \mathbf{u}) \geq -\alpha_1(h_1(\mathbf{x}))$$

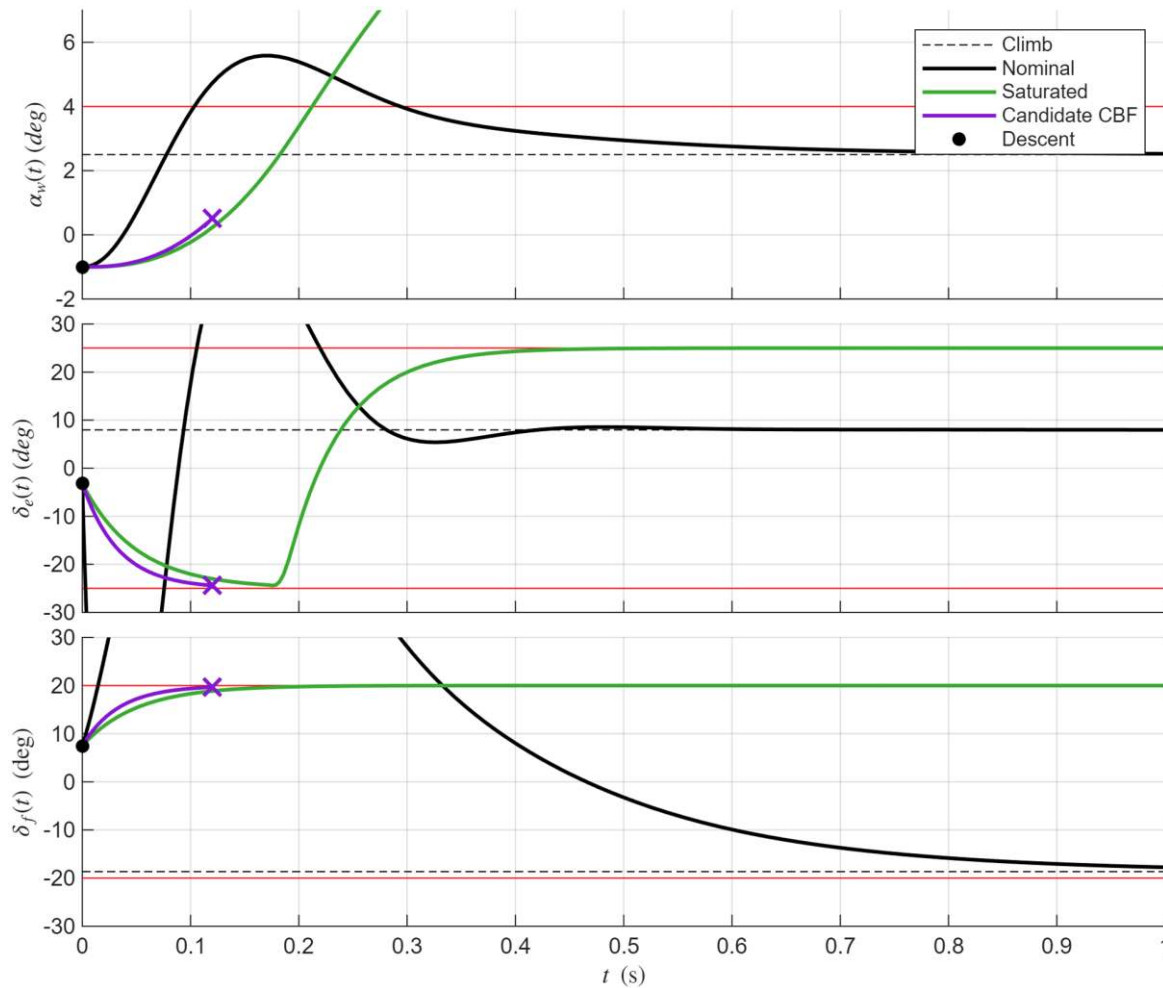
Infeasible



This procedure yields *candidate CBFs*.
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Example: Safety Filter for F-16



F-16 Pitch Dynamics

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$$\dot{h}_1(\mathbf{x}, \mathbf{u}) \geq -\alpha_1(h_1(\mathbf{x}))$$

Infeasible

Can be fixed by tuning α

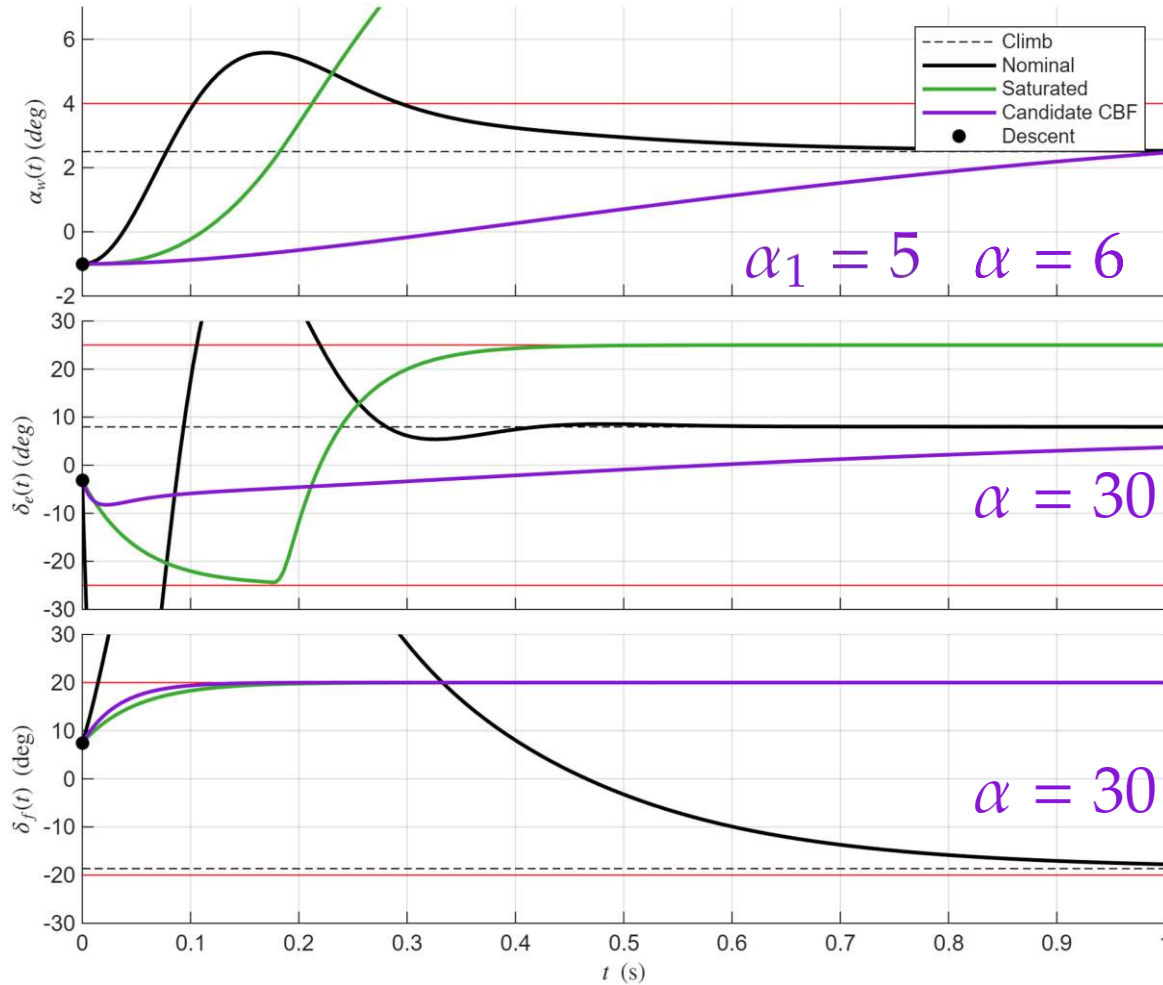


Example: Safety Filter for F-16

F-16 Pitch Dynamics

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$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$



$$\min_{\mathbf{u} \in \mathbb{R}^2} \|\mathbf{u} - \kappa(\mathbf{x})\|^2$$

$$\text{s.t.} \quad 20(\delta_e - \delta_{e,c}) \geq -30(25 - \delta_e)$$

$$20(\delta_f - \delta_{f,c}) \geq -30(20 - \delta_f)$$

$$\dot{h}_1(\mathbf{x}, \mathbf{u}) \geq -\alpha_1(h_1(\mathbf{x}))$$

Infeasible

Can be fixed by tuning α , bad performance

Index

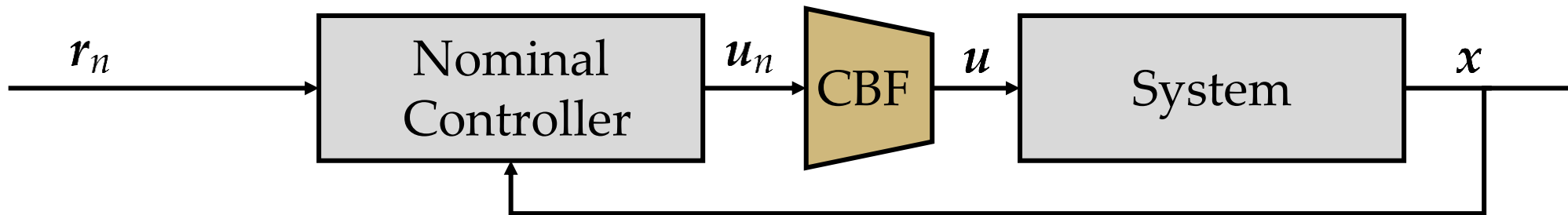
- ❖ Introduction
- ❖ Workshop Timeline
- ❖ Safety and Invariance
- ❖ Control Barrier Functions
- ❖ CBF-based Safety Filter
- ❖ Example: Safety Filter for F-16
- ❖ My Work: DSMs are CBFs



Safety Filters

Family of constrained controllers that use a two-step approach:

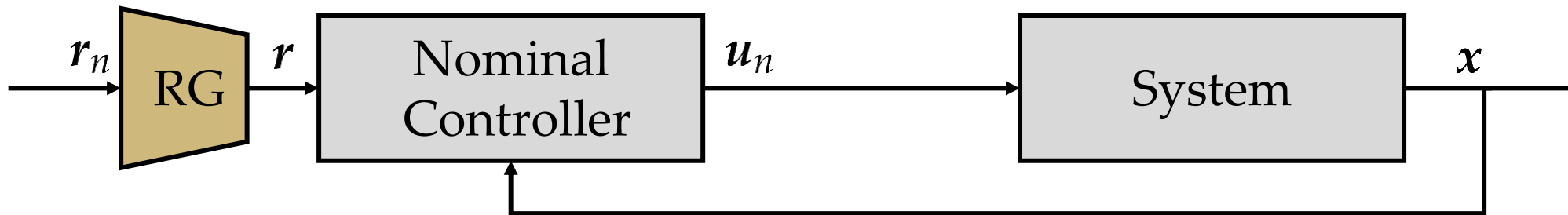
1. Design a control law for the **Unconstrained** system,
2. Introduce an **Add-on** unit for constraint enforcement.
 - **Control Barrier Function:** Filter nominal controller



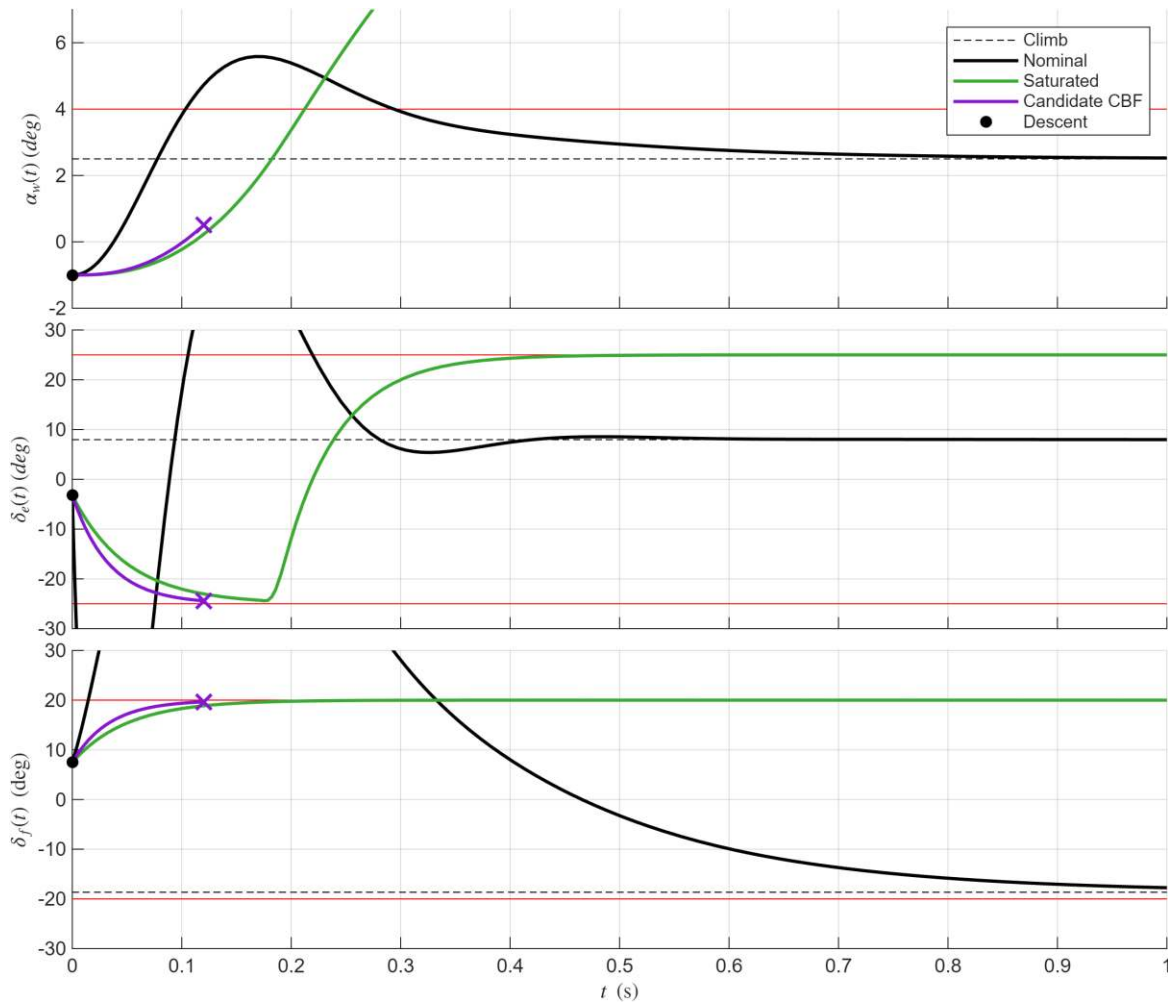
Safety Filters

Family of constrained controllers that use a two-step approach:

1. Design a control law for the **Unconstrained** system,
2. Introduce an **Add-on** unit for constraint enforcement.
 - **Control Barrier Function:** Filter nominal controller
 - **Reference Governor:** Filter nominal reference



My Work: DSMs are CBFs



F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

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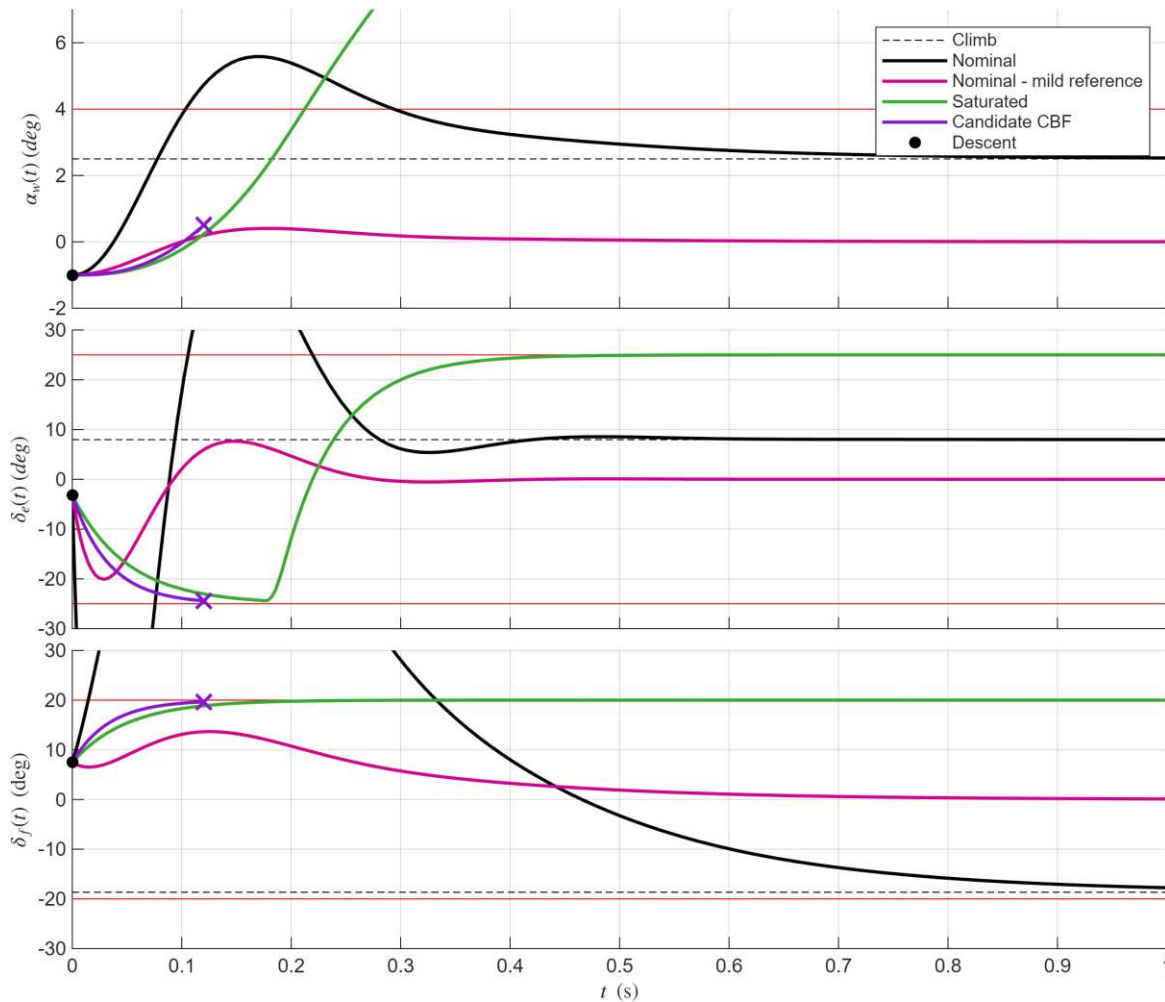
Descent: $\theta = -3, \gamma = -2$



Goal

Climb: $\theta = 9, \gamma = 6.5$

My Work: DSMs are CBFs



F-16 Pitch Dynamics

$$\mathbf{x} = \begin{bmatrix} \theta & \dot{\theta} & \alpha_w & \delta_e & \delta_f \end{bmatrix}$$

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Descent: $\theta = -3, \gamma = -2$

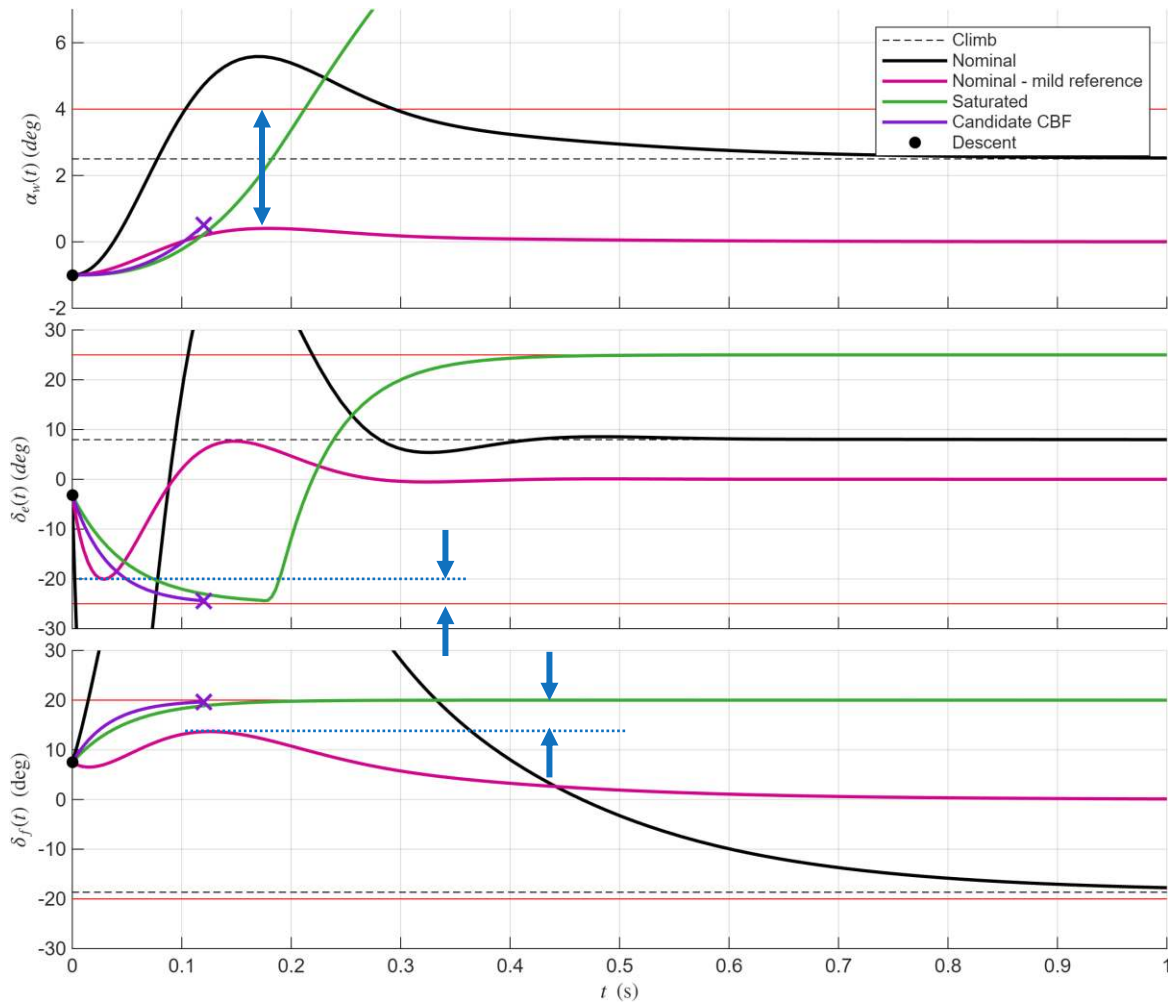


Intermediary Goal

Milder Descent: $\theta = -1, \gamma = -1$

LQR Controller $\kappa(\mathbf{x}) = \bar{\mathbf{u}}(\mathbf{v}) - \mathbf{K}(\mathbf{x} - \bar{\mathbf{x}}(\mathbf{v}))$

My Work: DSMs are CBFs



F-16 Pitch Dynamics

$$\mathbf{x} = \begin{bmatrix} \theta & \dot{\theta} & \alpha_w & \delta_e & \delta_f \end{bmatrix}$$

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Descent: $\theta = -3, \gamma = -2$

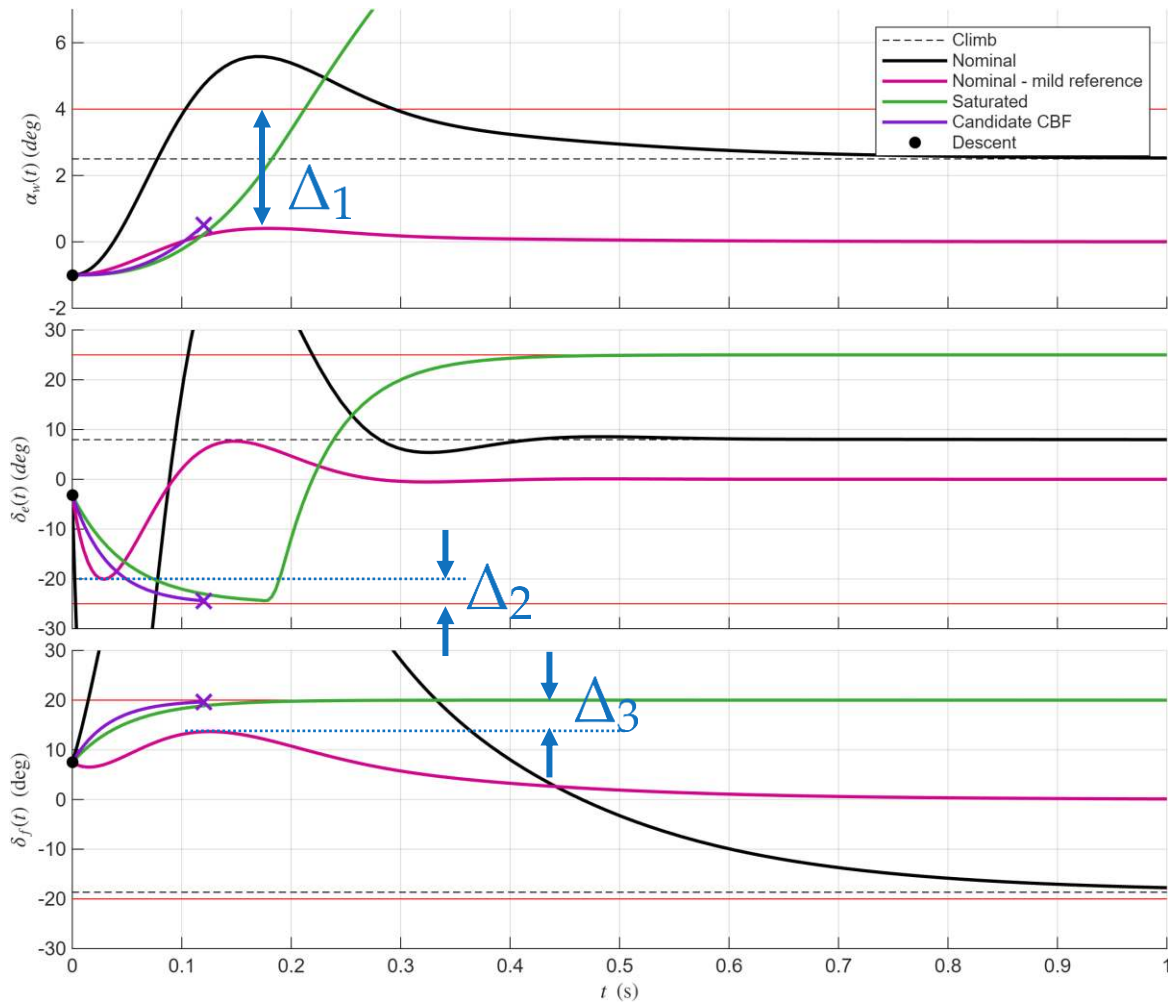


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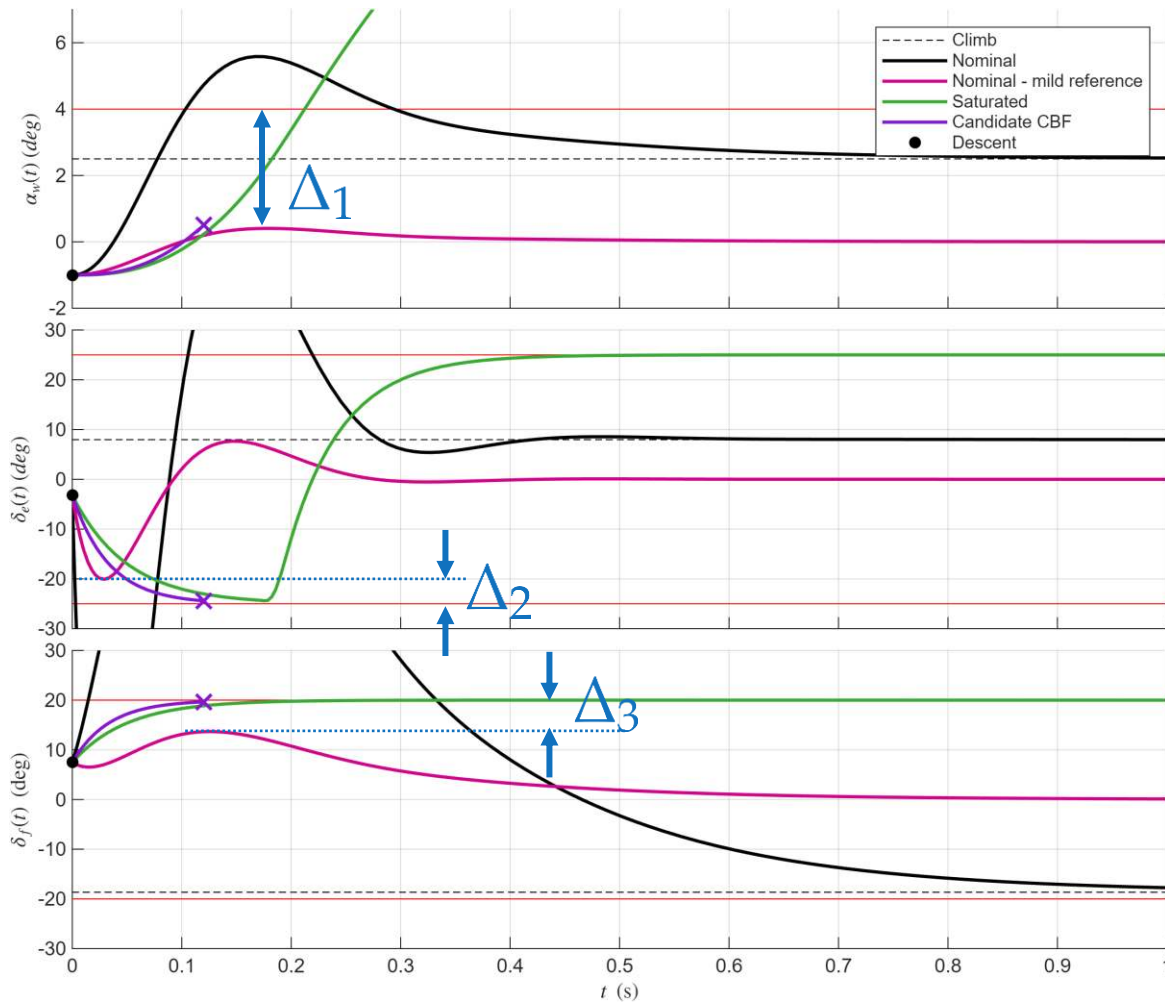


Dynamic Safety Margin (DSM)

$$\Delta(\mathbf{x}, \mathbf{v}) = \inf_{\tau \in [0, \infty)} h(\Phi(\tau, \mathbf{v}))$$

$$\text{LQR Controller } \kappa(\mathbf{x}) = \bar{\mathbf{u}}(\mathbf{v}) - K(\mathbf{x} - \bar{\mathbf{x}}(\mathbf{v}))$$

My Work: DSMs are CBFs



F-16 Pitch Dynamics

$$\mathbf{x} = \begin{bmatrix} \theta & \dot{\theta} & \alpha_w & \delta_e & \delta_f \end{bmatrix}$$

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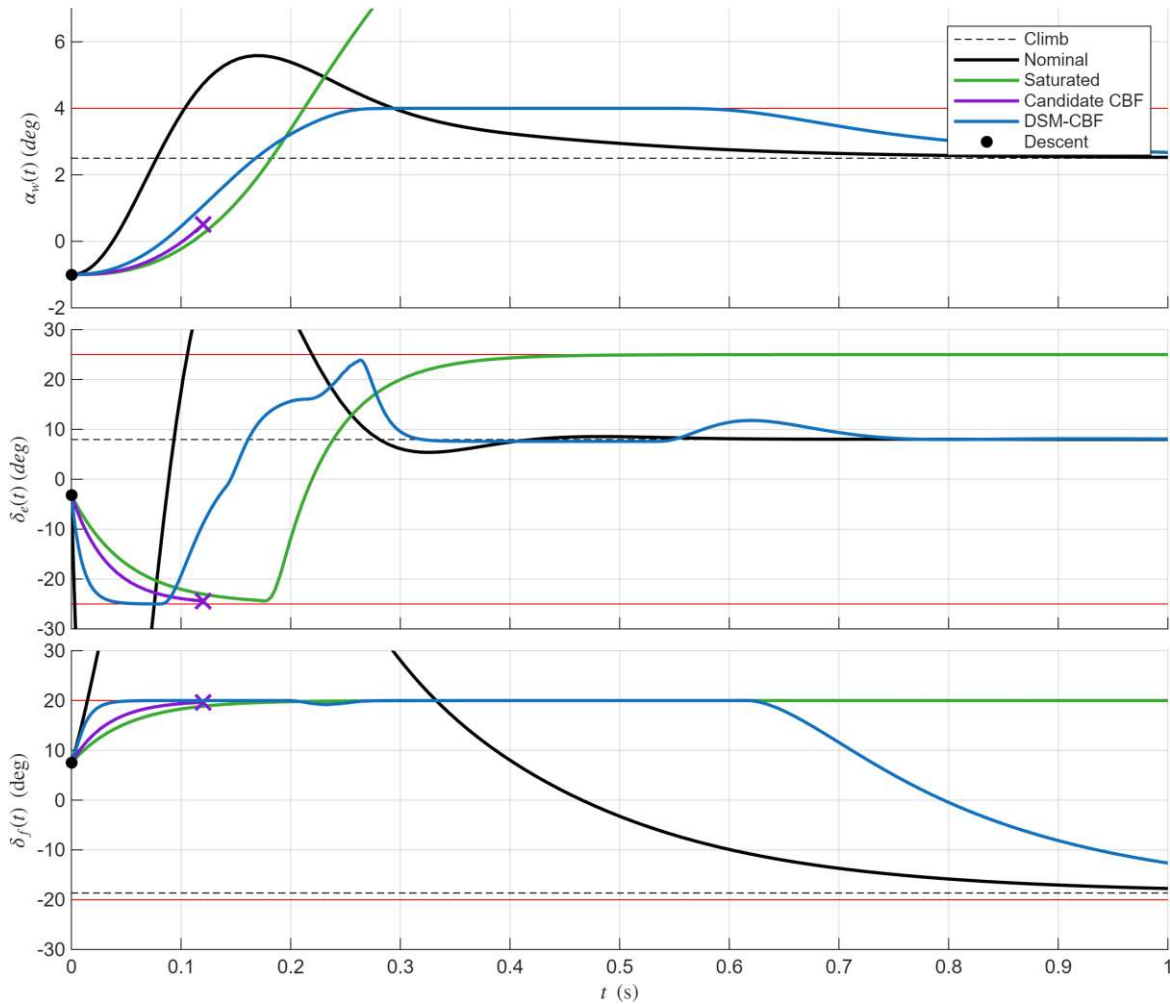


Dynamic Safety Margin (DSM)

$$\Delta(\mathbf{x}, \mathbf{v}) = \inf_{\tau \in [0, \infty)} h(\Phi(\tau, \mathbf{v}))$$

Theorem: DSMs are CBFs
 If Δ is a DSM, then it is a CBF for an augmented system $\begin{bmatrix} \mathbf{x} & \mathbf{v} \end{bmatrix}^\top$.

My Work: DSMs are CBFs



F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$

$$\min_{\mathbf{u} \in \mathbb{R}^2} \|\mathbf{u} - \kappa(\mathbf{x})\|^2$$

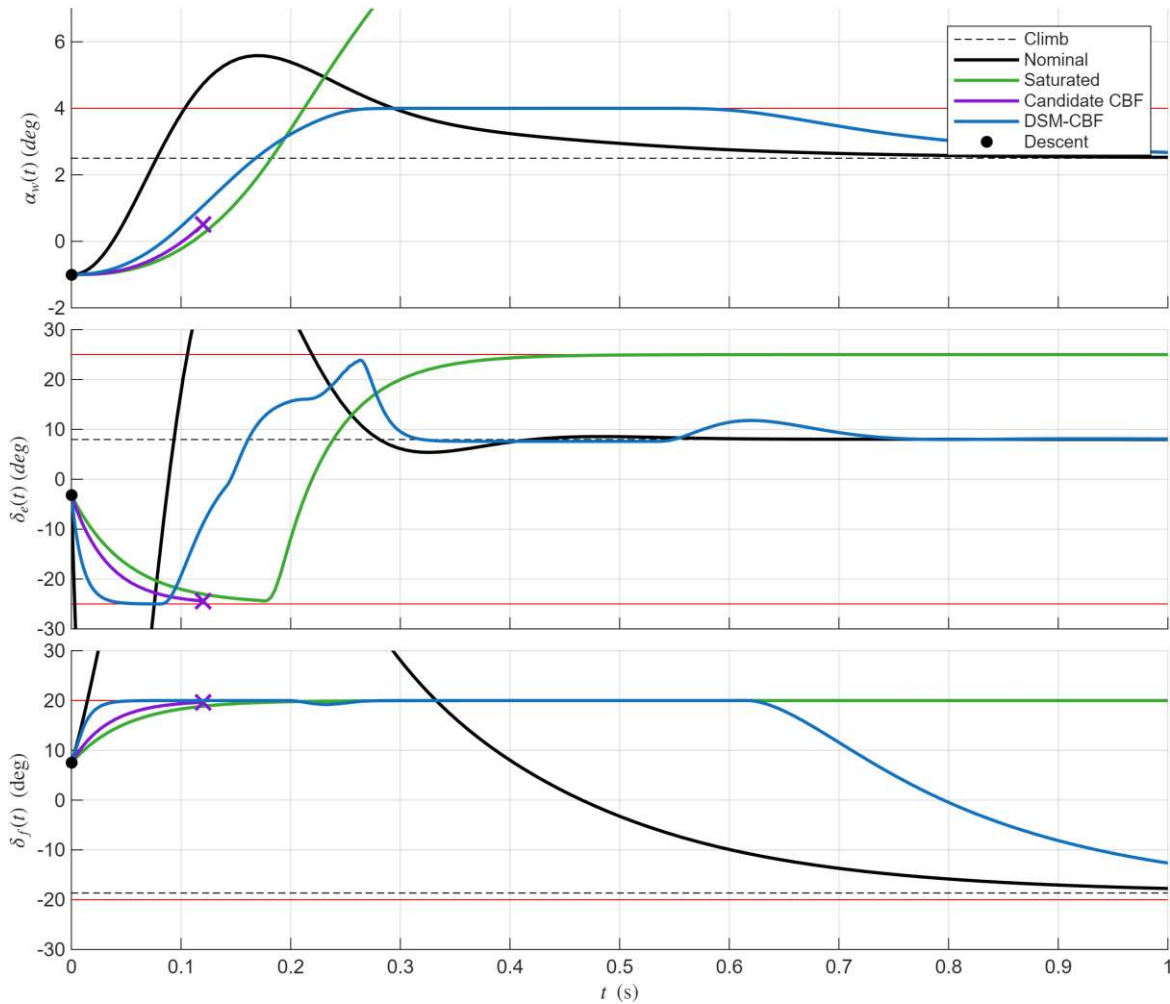
$$\text{s.t.} \quad \dot{\Delta}_1(\mathbf{x}, \mathbf{v}, \mathbf{u}) \geq -\alpha_1(\Delta_1(\mathbf{x}, \mathbf{b}))$$

$$\dot{\Delta}_2(\mathbf{x}, \mathbf{v}, \mathbf{u}) \geq -\alpha_2(\Delta_2(\mathbf{x}, \mathbf{b}))$$

$$\dot{\Delta}_3(\mathbf{x}, \mathbf{v}, \mathbf{u}) \geq -\alpha_3(\Delta_3(\mathbf{x}, \mathbf{b}))$$



My Work: DSMs are CBFs



F-16 Pitch Dynamics

$$\mathbf{x} = \begin{bmatrix} \theta & \dot{\theta} & \alpha_w & \delta_e & \delta_f \end{bmatrix}$$

$$\mathbf{u} = \begin{bmatrix} \delta_{e,c} & \delta_{f,c} \end{bmatrix}$$

$$\min_{\mathbf{u} \in \mathbb{R}^2} \|\mathbf{u} - \kappa(\mathbf{x})\|^2$$

$$\text{s.t. } \dot{\Delta}_1(\mathbf{x}, \mathbf{v}, \mathbf{u}) \geq -\alpha_1(\Delta_1(\mathbf{x}, \mathbf{b}))$$

$$\dot{\Delta}_2(\mathbf{x}, \mathbf{v}, \mathbf{u}) \geq -\alpha_2(\Delta_2(\mathbf{x}, \mathbf{b}))$$

$$\dot{\Delta}_3(\mathbf{x}, \mathbf{v}, \mathbf{u}) \geq -\alpha_3(\Delta_3(\mathbf{x}, \mathbf{b}))$$

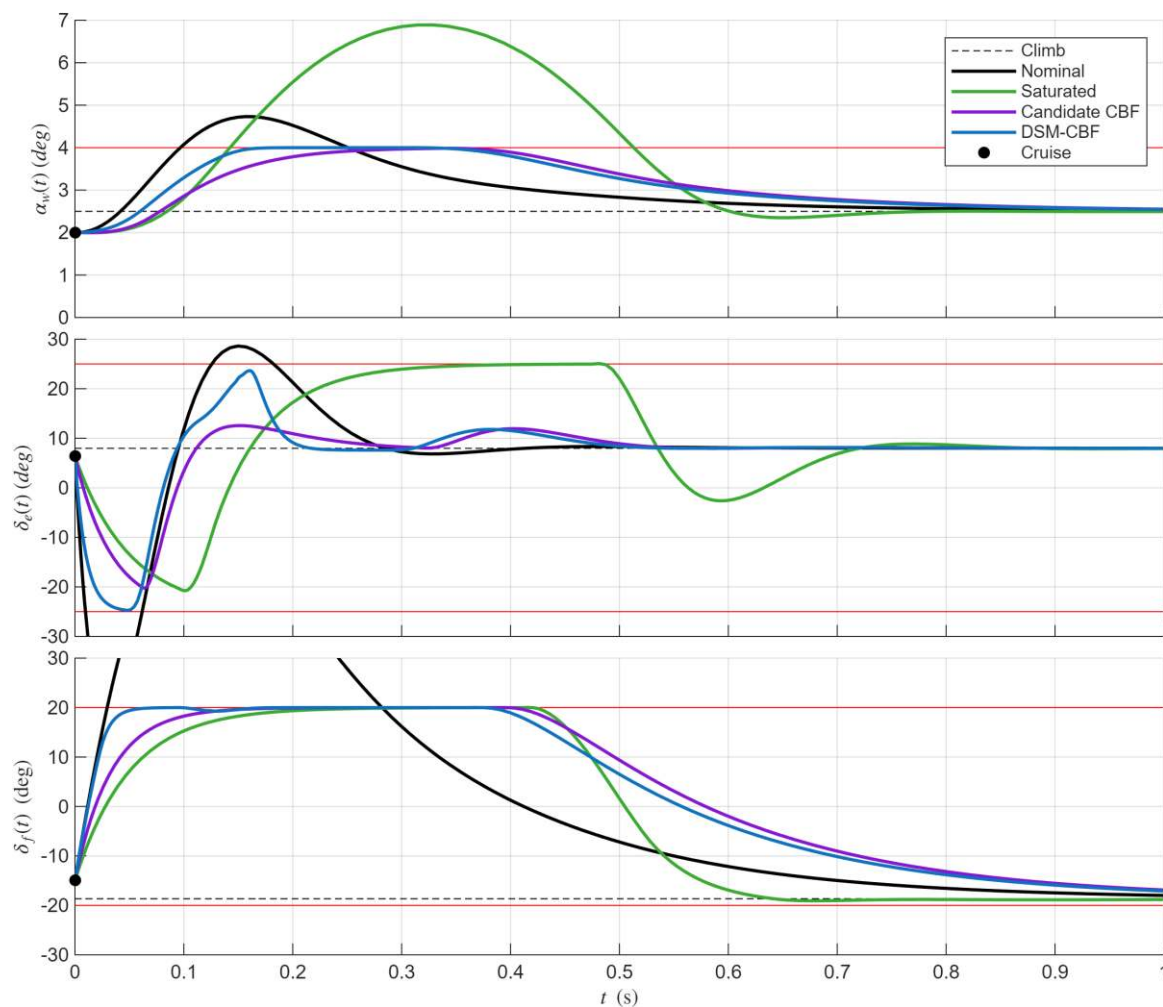


mosek

DSM-CBF: 16 ms solvetime

Candidate CBF: 3.5 ms solvetime

My Work: DSMs are CBFs



F-16 Pitch Dynamics

$$\mathbf{x} = [\theta \quad \dot{\theta} \quad \alpha_w \quad \delta_e \quad \delta_f]$$

$$\mathbf{u} = [\delta_{e,c} \quad \delta_{f,c}]$$



Cruise: $\theta = 2, \gamma = 0$



Goal

Climb: $\theta = 9, \gamma = 6.5$

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Candidate CBF: 3.5 ms solvetime

Acknowledgements

